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REVIEW

Soft Computing Techniques for Biometric Personal Authentication: Fundamentals, Challenges, and Open Issues

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Abstract

Biometric authentication systems are increasingly deployed across security-critical domains due to their ability to verify identity using permanent and unforgeable physiological or behavioral traits. However, single-modality (unimodal) biometric approaches remain susceptible to noise, intra-class variation, spoofing, and failure-to-enrol problems. Soft computing techniques—encompassing fuzzy logic, artificial neural networks, evolutionary algorithms, swarm intelligence, and deep learning—offer adaptive, uncertainty-tolerant solutions that are particularly well-suited to the complex, high-dimensional nature of biometric data. This review systematically examines the application of soft computing methodologies to the biometric personal authentication, with emphasis on multimodal systems that integrate multiple biometric sources through sensor-level, feature-level, score-level, rank-level, and decision-level fusion strategies. Following a PRISMA 2020-compliant screening procedure across Scopus and Web of Science (2010–2024), a total of 153 peer-reviewed studies are synthesized. The review provides a structured comparative analysis of soft computing paradigms across four operationally relevant dimensions—recognition accuracy, computational cost, scalability, and robustness to noise—and introduces an original taxonomy categorizing methods by their architectural role in the biometric pipeline. Six open challenges are identified: cross-sensor generalization, adversarial robustness, privacy and ethical concerns, IoT and edge deployment constraints, model explainability, and demographic bias. Five specific future research directions are outlined, including explainable AI integration, federated learning for privacy-preserving multimodal training, and lightweight hybrid architectures for real-time IoT deployment.

1 Introduction

Biometric systems are becoming more and more common due to the growing need for security and identification solutions. Multibiometric systems are being developed for several reasons. One of the main reasons is that a person might conceal or not have the biometrics on which the system is built. 'Biometrics' refers to the automated process of recognizing individuals using their physical or behavioral traits, such as facial features, voice, iris patterns, gait, or signature [1]. These characteristics are frequently called biometric cues or biometric modalities. Many biometric modalities [2, 3] have been investigated in the last few years for usage in a variety of applications, from border control systems to personal device access systems. Biometric authentication is increasingly being adopted across various security applications to confirm an individual's identity. Systems that depend on the physiological characteristics of their users have recently been able to be more reliable and simpler at the same time. This has assisted in preventing a number of issues related to systems that employ passwords or credentials, which may result in issues like lost, transferred, or stolen credentials. Because a person with certain biometric characteristics must authenticate access to the systems in order to prevent the aforementioned issues, the adoption of biometrics has greatly reduced these concerns.

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Soft computing techniques use an approximate solution for issues for which there is no known algorithm for calculating the exact solution. For a number of reasons, soft computing has become more prevalent in biometrics. The primary one is that biometric data is inherently ambiguous and imprecise, raising the rates of false rejection and false acceptance (FAR and FRR, respectively). Soft computing is an extremely effective method for handling ambiguous and imprecise data.

1.1 Biometric Domain

Biometrics involves recognizing individuals through their physical or behavioral characteristics [4]. In recent years, its application in authentication systems has seen significant growth. This is largely because biometric features, such as a person's fingerprint, face, and iris, are permanent and difficult to guess, forget, or misplace. It has made great strides in a number of fields, including security, surveillance, access control, and authentication. In contrast to conventional forms of authentication, such as tokens or passwords, biometric characteristics are difficult to falsify and are not easily forgotten.

Figure 1 summarizes the general architecture of a typical biometric recognition system. Here, a standard biometric recognition system typically begins with a segmentation or detection phase, where the relevant biometric modality is isolated from the raw input—such as an image, video stream, or physiological signal. This is followed by a preprocessing stage, which may involve tasks such as data alignment, denoising, or augmentation to enhance the quality and consistency of the extracted information. A classifier for biometric recognition uses the features that have been derived from the preprocessed data. The recognition process in biometrics typically involves either determining the identity associated with a given input, commonly referred to as biometric identification, or assessing whether two input samples correspond to the same individual, known as biometric verification.

Unimodal biometric systems, which depend on a single modality, suffer from several inherent limitations, including intra-class variation, vulnerability to spoofing attacks, and high failure-to-enroll rates [5]. Multibiometrics is a viable solution to get over this restriction since it allows us to use multiple modalities simultaneously to increase system accuracy and performance. In multibiometrics, a technique called fusion is crucial in combining the characteristics of

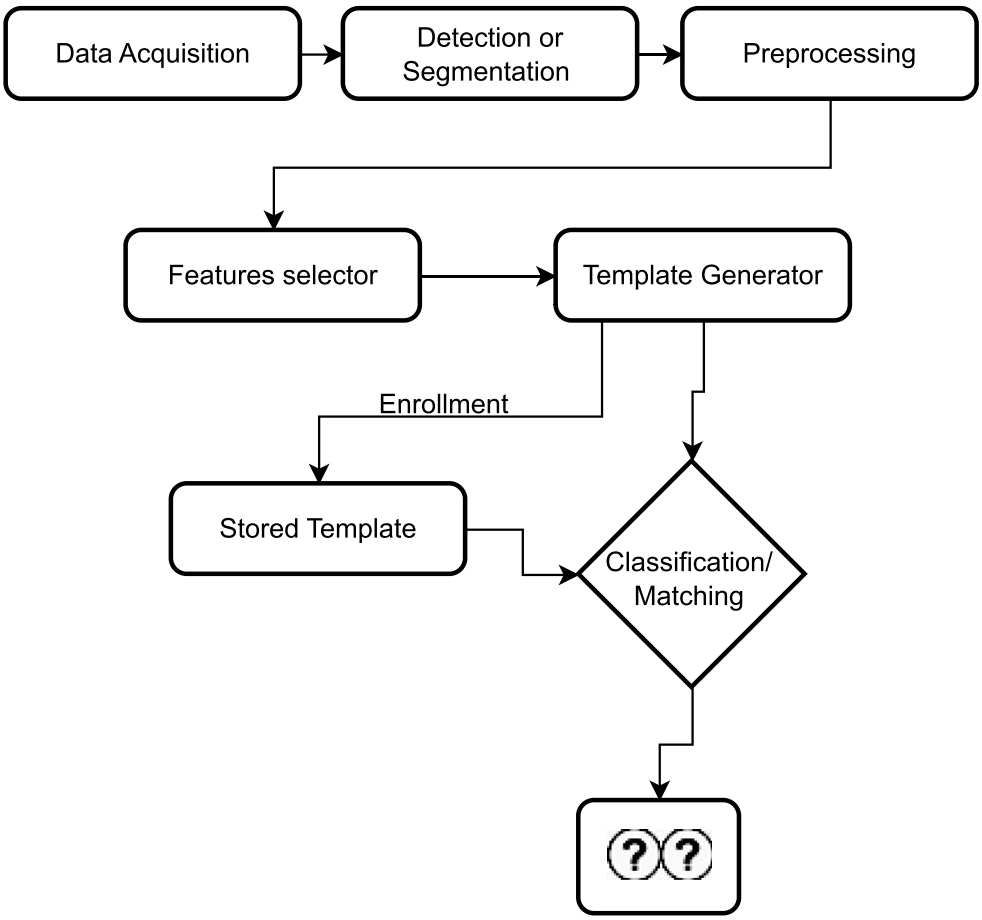


Fig. 1 Design structure of biometric recognition systems

many modalities [1]. The integration of numerous sources of biometric data through a multibiometric recognition system is a crucial component to enhancing the recognition performance of biometric-based systems. To improve classification performance, it seeks to incorporate the identity evidence provided by many sources of biometric data [6]. When such systems are implemented, the performance of biometric systems tends to improve because more discriminating information that is accessible for each individual is used. Multibiometric systems offer several advantages compared to unimodal biometric systems. including the ability to: (1) significantly increase the accuracy of the biometric authentication process; (2) offer a high level of flexibility by compensating for unusable or unfavorable biometric traits in specific individuals with data from other biometric modalities; and (3) reduce the risk of biometric fraud, and (4) add an additional layer of difficulty to prevent spoofing assaults.

Because the feature representation of many qualities may differ and be incompatible with one another, multibiometric feature fusion is not simple. Based on the nature of biometric data sources, multibiometric systems can be classified into six categories: multimodal, multi-algorithm, multi-sample, hybrid, multi-sensor, and multi-instance [6], as illustrated in Fig. 2.

Multi-sensor systems integrate data obtained from various sensors for a single biometric modality. For instance, depth information from a 3D camera [8], and infrared data from an NIR camera could all be used by a face recognition module [9, 10]. An algorithm for multi-sensor fusion would be produced if a subject was identified using both images. These systems use several sensors to obtain various information from a single modality, which they rely on for recognition [3]. Multi-sensor systems are in handy when different modes of capture are needed at different times or when different sensors can successfully capture discriminative information.

Multiple algorithms are used by multi-algorithm systems to process an input sample. In many biometric systems, a single sensor is employed to capture data from a specific modality; however, multiple algorithms may be applied to process and interpret that data. As an example, palmprint recognition can be performed using appearance-based representations to facilitate matching, while a fingerprint recognition system might integrate both texture features and minutiae-based information to improve identification accuracy [11]. Different kinds of information can be extracted and used from the same sample. Multi-algorithm systems often yield improved performance when the combined algorithms or feature sets provide complementary information.

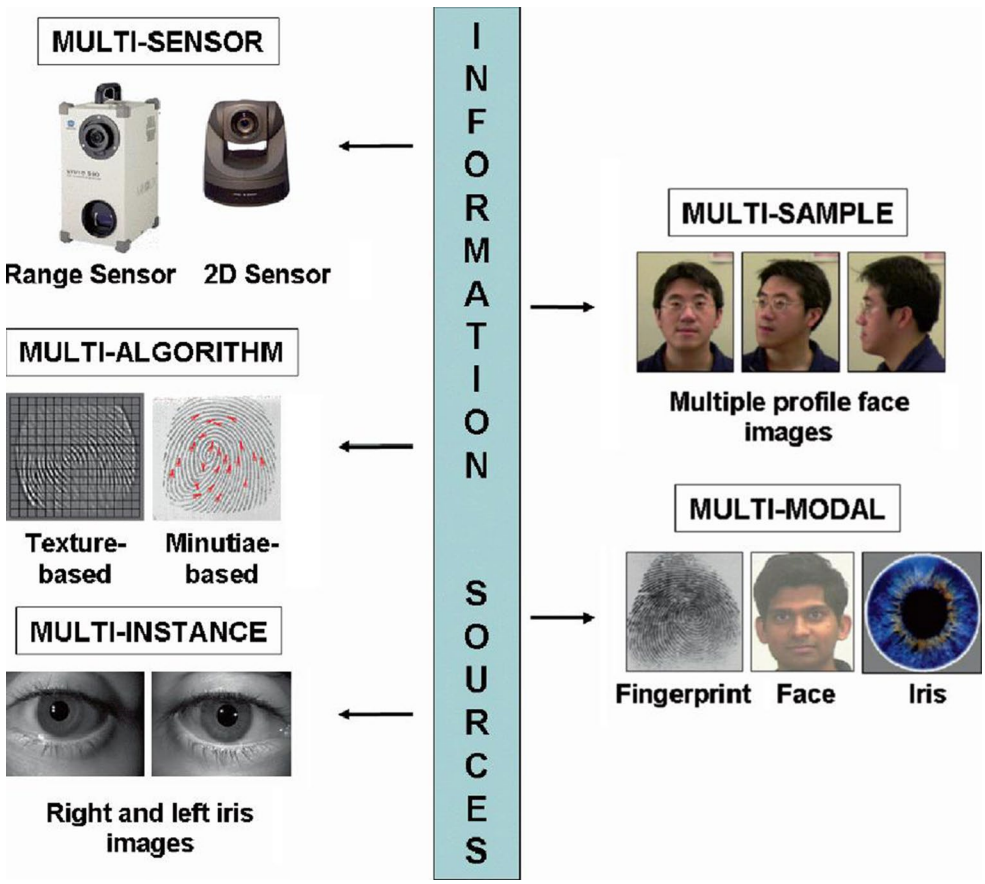


Fig. 2 Types of information used in biometric fusion [7]

The same biometric attribute can be captured by several instances using **multi-instance systems**. In iris recognition, the system can utilize both the left and right irises, forming a multi-instance approach. Similarly, multi-instance biometric systems leverage data from multiple samples of the same biometric modality, for example, all ten fingers or both palms in fingerprint or palmprint identification [12, 13]. The feature extraction and matching methods used in single-instance approaches may frequently be applied consistently to all instances of the biometric trait in such systems.

Several samples of the same biometric modality, frequently obtained with considerable differences, are used in **multi-sample systems**. This category includes video-based recognition models, in which a biometric modality is constantly recorded for a brief amount of time (a few seconds, for example). As a result, there are frequently many frames with various types of information. Videos have been used in the literature to conduct gait [14, 15] and face [16, 17] identification. As a result, information from several video frames is combined to create a multi-sample recognition system. With just one sensor needed, these systems can extract a variety of data from a single biometric modality.

To identify a subject, **multi-modal systems** use information gathered from several biometric cues. A multimodal biometric system can integrate information from various combinations of biometric modalities, such as face, fingerprint, and iris; face, fingerprint, and speech; or face and voice [17, 18]. Other common modality pairs include ear and face, as well as iris and periocular regions [19, 20]. To perform speaker recognition, research has also focused on integrating other modalities, For example, combinations may include audio with lip motion [21], or a fusion of audio, lip motion, and lip texture. Due to their ability to process numerous biometric modalities, these systems help mitigate the shortcomings inherent in unimodal biometric approaches [22]. When a person can provide data for one biometric modality but not for another, multimodal systems can also be helpful [23]. Combining data from several modalities makes it possible to extract even more distinguishing features, which frequently leads to better identification performance.

A distinct type of multibiometric system combines core biometric modalities, such as face and fingerprint, with soft traits including gender, weight, eye color, and height. Since multiple members of the target population are likely to share the same soft biometric feature, it is impossible to properly differentiate between individuals using these traits. However, the effectiveness of the authentication process can be significantly enhanced when soft biometric traits are combined with primary biometric features [24]. Additionally, soft biometric attributes can aid in narrowing down large biometric databases by reducing the number of potential matches to be examined.

Given the constrained availability of biometric data, fusion in multimodal biometric systems can be performed at various stages, including the sensor (raw data) level, feature-level, matching score-level, and decision-level [5, 25]. However, because the feature set used by different biometric modalities is either inaccessible or incompatible, it becomes impossible to integrate information at the feature-level [26]. Consequently, due to the ease of consolidating and obtaining matching scores, the matching score is the generally recommended fusion level.

Sensor-level fusion, also referred to as data-level fusion, is typically associated with multisensor or multisample algorithms, in which data are fused as soon as they are acquired. At its core, raw data-level fusion is performed before the feature extraction stage. For example, facial images taken from various angles, such as front, right, and left profiles, can be fused using mosaicing techniques to create a comprehensive facial representation. Alternatively, direct fusion methods, including the addition of pixel levels from multiple images, are frequently employed for this purpose [27, 28].

Feature-level fusion refers to the process of consolidating feature vectors extracted from the same biometric source or multiple biometric modalities to create a richer and more discriminative combined representation for subsequent matching or classification [29]. It can also involve matching features extracted from multiple biometric modalities. These techniques are commonly utilized in multibiometric cryptosystems, which integrate features from multiple biometric modalities to enhance both security and user privacy [30]. Multimodal biometric database indexing has also been done using them [31].

Score-level fusion refers to the process of integrating the matching scores generated by multiple biometric matchers. Typical techniques used at this level include mean score fusion, maximum score fusion, and minimum score fusion, where the final decision score is derived by computing the average, maximum, or minimum value of the individual match scores [5, 32]. Advanced fusion techniques, including probabilistic approaches such as likelihood ratio-based score fusion, as well as evidence-based methods like Dempster–Shafer theory, have also been employed in multibiometric systems [33]. Furthermore, Ding and Ross address several imputation methods for managing incomplete or missing data in the context of score-level fusion [34]. Because it is so simple to obtain the scores produced by commercial matchers.

Rank-level fusion is performed after the probe sample is compared against the template set within the gallery or database. In identification tasks, the matcher typically generates

a sorted list of matched identities by comparing a supplied probing image against a gallery of photos. Several methods, including the Borda count, logistic regression, and the highest-rank approach, have been employed in the literature to merge rank lists generated by multiple biometric matchers [35, 36].

There are two primary frameworks for classifier combinations: Within the first framework, all classifiers utilize a uniform representation of the input modalities. In the second framework, every classifier uses a different presentation of the input modality, and every classifier has its own measurement that is taken from the modality [37, 38].

Sequential and parallel fusion models are the two categories. In parallel fusion, user identification is achieved by sequentially evaluating individuals across multiple biometric modalities, where each phase admits a subset of subjects based on a single modality. If a subject is successfully authenticated at any given phase, subsequent stages are bypassed. As a result, fewer biometric modalities are needed overall for subject identification, and efficiency is increased by cutting down on processing time. Sequential fusion, on the other hand, uses several biometrics at once; consumers are first introduced to one biometric modality before moving on to the next [39–41]. Since all biometric modalities are accepted for each user's verification, the parallel fusion offers greater accuracy.

1.2 Soft Computing Overview

Soft computing encompasses a family of computational paradigms designed to build intelligent systems that emulate human reasoning under conditions of imprecision, uncertainty, and partial truth [42]. Neural networks, fuzzy logic, and evolutionary computing are examples of the different computational models that make up soft computing. For the system's intelligent information processing, these models collaborate. For a variety of reasons, {systems that integrate multiple soft computing paradigms are particularly well-suited to biometric recognition, owing to their ability to handle ambiguous, high-dimensional, and noisy biometric data. One of the fundamental challenges in biometric systems lies in ensuring that biometric traits are invariant to changes in conditions or environments, even though such traits may naturally vary over time. Soft computing techniques offer adaptability to these variations, enhancing system robustness. In multimodal biometrics, multiple features from different modalities are integrated to produce a final decision. The fusion process may be implemented at multiple stages, including sensor-level, feature-level, score-level, and rank-level integration. The ability to fuse several traits is a key component of multimodal biometrics; nevertheless, the complexity of the fusion is raised since features retrieved from one person's biometric trait are independent of those extracted from another. Because soft computing techniques can manage ambiguous and incompatible data, they greatly aid in minimizing the fusion complexity. Feature extraction is a critical component of biometric identification systems, wherein distinctive traits are quantified and converted into a mathematical representation referred to as a biometric template. Variant data can be handled very effectively with soft computing approaches [43, 44]. Figure 3 illustrates the several applications of soft computing in biometrics.

Soft computing heavily relies on the artificial intelligence subfield of machine learning. Without explicit programming, it enables systems to improve over time by learning from experience [45]. ML algorithms use facts, observations, or prior experiences to generate defensible conclusions.

Deep learning, a branch of machine learning within the broader domain of soft computing, utilizes multi-layered artificial neural networks to learn hierarchical representations of data. These models are capable of automatically extracting complex features directly from raw input, thereby eliminating the need for manual feature engineering [46]. Soft computing is a general term for a range of computational methods inspired by human-like intelligence that are used to solve difficult and unpredictable situations. Soft computing encompasses a range of methodologies, including neural networks, fuzzy logic, probabilistic reasoning, and evolutionary algorithms. Deep learning aligns closely with soft computing principles, owing to its neural network-based architecture and its ability to learn from vast and complex datasets.

Deep learning has demonstrated transformative potential across domains including natural language processing, speech recognition, and computer vision. Its capacity to automatically extract hierarchical representations from large-scale data has driven significant advances in industries ranging from financial services and healthcare to autonomous systems. Soft computing techniques, such as deep learning, make sophisticated solutions for difficult data processing, pattern recognition, and decision-making problems possible by utilizing deep neural networks. Many applications' accuracy and performance have been greatly enhanced by their capacity to handle unstructured input and learn from mistakes.

1. **Fuzzy Logic:** Rather than dealing with precise and fixed values, fuzzy logic is a many-valued logic that works with approximative reasoning. It is made up of membership functions, fuzzy sets, and a collection of rules for handling different computational issues. Biometric uncertainty issues have been addressed by fuzzy inference systems that use fuzzy logic because accurate biometric data is rarely obtained. Additionally, fuzzy systems are

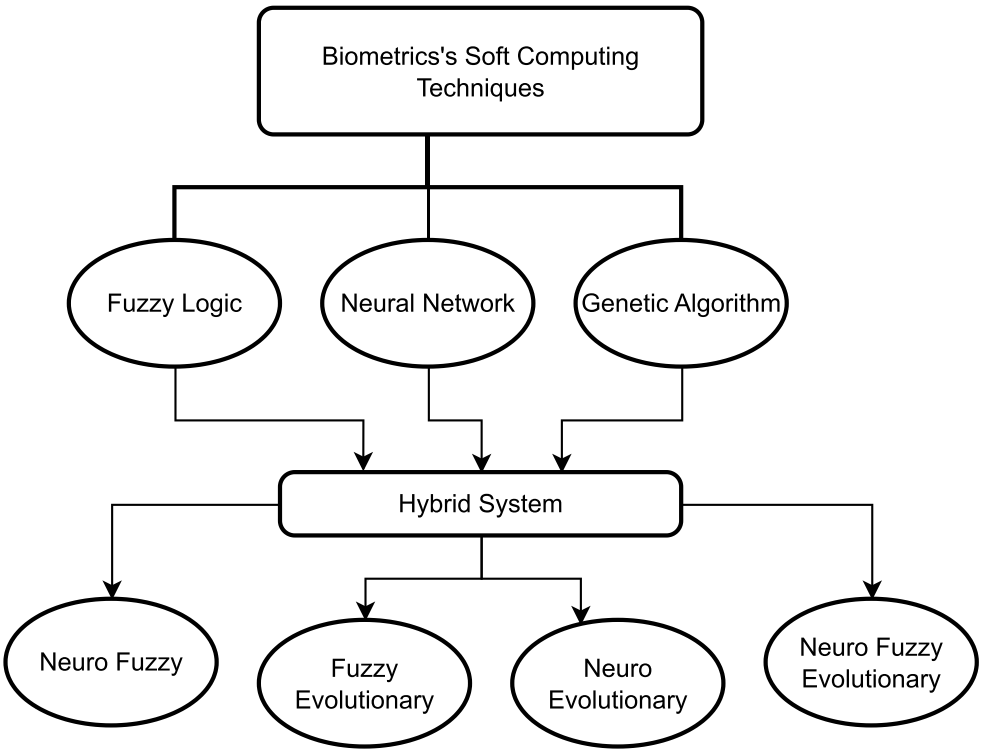


Fig. 3 Proposed taxonomy of soft computing techniques in biometric authentication, categorized by paradigm (standalone, hybrid, and deep-integrated architectures) and their primary role in the biometric pipeline (feature extraction, feature selection, classification, and fusion)

employed in multimodal biometrics to enhance human recognition rates and carry out different forms of fusion [44, 47]. Multimodal biometric systems can incorporate various fuzzy logic-based approaches, including fuzzy decision-making systems, fuzzy fusion techniques, and fuzzy binary decision trees [48].

2. **Artificial Neural Networks (ANN):** An ANN consists of a network of interconnected processing units, known as neurons, inspired by the structure and functioning of the human brain. Each neuron performs a mathematical operation that aggregates weighted inputs and applies an activation function to produce the corresponding output. Neural networks have two key characteristics that make them useful for biometric applications: learning and adaptability [49, 50]. Applications of neural networks in biometrics include person recognition, feature extraction, preprocessing, and interpretation [51, 52]. By conjugating more than one neural network, the system's efficiency can be further boosted [53]. There are two techniques to join neural networks: ensemble-based neural networks and modular neural networks.
3. **Evolutionary Algorithms:** Natural selection is the foundation of evolutionary algorithms, which are metaheuristic optimization techniques. To carry out optimization, evolutionary algorithms make use of crossover, mutation, and domain-specific operators [54]. Evolutionary algorithms have applications in biometrics where they can be used to choose the optimum feature set, improve the biometric databases, and reduce ambiguity [55, 55]. In biometrics, matching can also be done by evolutionary algorithms [44]. Fuzzy logic and neural networks exhibit enhanced performance and improved efficiency in executing tasks when integrated with evolutionary algorithms [56, 57].

1.3 Motivation

The rapid proliferation of digital systems and the increasing demand for secure, reliable, and user-friendly authentication methods have driven the widespread adoption of biometric systems. However, traditional unimodal biometric approaches often struggle with issues such as noise in the data, intra-class variations, spoofing vulnerabilities, and failure-to-enroll problems. These challenges have led to a growing interest in more robust and adaptive frameworks, with soft computing techniques being among the most prominent.

Soft computing encompasses a suite of computational methods, such as fuzzy logic, neural networks, evolutionary algorithms, and swarm intelligence, all of which are inherently designed to handle imprecision, partial truths, and uncertainty. This makes them particularly well-suited for processing complex, high-dimensional, and sometimes noisy biometric data. Unlike conventional algorithms that require rigid, rule-based logic, soft computing methods can learn from data, adapt over time, and provide flexible decision-making capabilities, which are essential in dynamic and heterogeneous biometric environments.

Moreover, the integration of soft computing with biometric systems enables the development of advanced multimodal authentication schemes. These systems combine information from multiple biometric sources (e.g., face, fingerprint, iris, voice), thereby enhancing accuracy, reducing error rates, and increasing resilience against spoofing attacks. Soft computing techniques further facilitate effective feature selection, dimensionality reduction, classification, and fusion strategies, ultimately leading to more intelligent and context-aware biometric authentication systems.

In summary, the motivation to explore soft computing techniques in biometric authentication stems from their ability to overcome the shortcomings of conventional systems and their potential to enable scalable, secure, and adaptive identity verification solutions. This review aims to critically analyze the contributions of soft computing methods in this domain, highlighting their strengths, limitations, and prospects.

1.4 Contribution

This review provides a structured and critically oriented examination of soft computing methodologies in biometric personal authentication, integrating methodological coverage with comparative analysis, application context, and challenge identification. The principal contributions are outlined as follows:

1. **Integrated Review of the Soft Computing Spectrum in Biometrics:** A unified examination is provided of the full range of soft computing paradigms applied to biometric authentication—encompassing fuzzy logic, artificial neural networks, evolutionary algorithms, swarm intelligence, and deep learning architectures. For each paradigm, the analysis focuses on its specific functional role within the biometric pipeline: feature extraction, classification, fusion, or optimization, providing domain-specific rather than generic computational perspective.
2. **Structured Comparative Analysis and Critical Synthesis:** A structured comparative analysis evaluates nine soft computing methods and hybrid architectures across four operationally relevant dimensions: recognition accuracy, computational cost, scalability, and robustness to noise. Beyond tabular comparison, a critical synthesis identifies consistent performance patterns across reviewed studies—explaining, for instance, why evolutionary algorithms demonstrate superior feature selection efficacy, why fuzzy-neural hybrid systems outperform standalone classifiers in decision-level fusion, and why deep learning models generalize poorly across heterogeneous sensors despite achieving high benchmark accuracy. This analytical layer distinguishes the present work from purely descriptive surveys.
3. **Proposed Taxonomy of Soft Computing Roles in Biometric Systems:** An original taxonomy is introduced that categorizes soft computing techniques according to their architectural integration in biometric systems: standalone methods, hybrid architectures, and deep-integrated frameworks. This taxonomy further distinguishes methods by their primary pipeline function—feature extraction, feature selection, classification, and fusion—providing a structured reference for system designers.
4. **Systematic Analysis of Multimodal Fusion Strategies:** Fusion in multibiometric systems is examined across all integration levels—sensor, feature, score, rank, and decision—with emphasis on how soft computing techniques address the fundamental challenge of each level, including feature incompatibility, score normalization uncertainty, and decision-level conflict resolution. Both parallel and sequential fusion architectures are discussed in terms of their accuracy–efficiency trade-offs.
5. **Application-Driven Synthesis Across Deployment Domains:** Soft computing applications in biometric authentication are reviewed across six real-world deployment contexts: smartphone authentication, surveillance and security systems, online access control, financial and governmental services, IoT security, and hybrid cryptobiometric frameworks. This application-centred perspective connects methodological choices to operational requirements, providing evidence-based guidance grounded in reviewed experimental results.
6. **Structured Identification of Open Challenges and Future Research Priorities:** Open challenges are structured into six categories: data variability and sensor noise, adversarial robustness, privacy and ethical considerations, IoT and edge deployment constraints, model explainability, and cross-sensor generalization. For each challenge, candidate soft computing solutions are identified, and specific future research directions are outlined, providing an actionable roadmap for the field.

While several related surveys address aspects of this domain, they differ from the present review in scope and focus. Minaee et al. [58] provide a comprehensive survey of deep learning for biometric recognition; however, they do not address fuzzy logic, evolutionary algorithms, or swarm intelligence, nor do they employ a systematic PRISMA-based methodology. Pahuja and Goel [59] review multimodal biometric authentication at the algorithmic level but do not provide a structured soft computing taxonomy or challenge-to-solution mapping. Houssein et al. [60], while adopting a systematic methodology, focus on medical imaging rather than biometric authentication.

The present review is therefore positioned as complementary to these surveys, specifically addressing the intersection of the full spectrum of soft computing techniques and multimodal biometric system design—an area that remains insufficiently synthesized in the current literature. A structured comparison of these surveys is provided in Table 1.

As shown in Table 1, the present review differs from existing surveys by integrating the full spectrum of soft computing techniques, including fuzzy logic, evolutionary algorithms, swarm intelligence, machine learning, and deep learning, within a single biometric authentication framework. In addition, the review combines a PRISMA-compliant methodology, structured comparative analysis, and a taxonomy-driven perspective on the roles of soft computing techniques across biometric pipeline.

1.5 Review Structure

The remainder of this review is organized as follows: Section 2 provides a comprehensive overview of the methodology used to gather pertinent research data, encompassing the development of research questions, keyword selection, and criteria for including studies. An extensive review of the background data and underlying ideas for soft computing techniques, different multibiometric databases, and assessment metrics is given in Section 3 of the work. It provides as a framework for comprehending the portions that follow. Section 4 focuses on the topic of soft computing applications, namely machine learning in biometrics domain. The use of meta-heuristic approaches in this setting is also covered in this section. Section 5 integrates the findings and observations from prior sections, offering a comparative evaluation of different soft computing techniques and their specific applications within biometric systems. This section provides a thorough analysis of the advantages, disadvantages, and relative effectiveness of these approaches. Future directions and several difficulties in the area of soft computing in biometrics are examined in Sect. 6. The section highlights areas that need more research and development and highlights the constraints and challenges that practitioners and researchers may run into when putting these strategies into practice. Section 7 provides a concluding summary of the review, highlighting the major discoveries and contributions covered throughout the work. It offers a cogent wrap-up to the review, highlighting the importance of the examined soft computing methods in biometrics domain.

Table 1 Structured comparison of the present review with closely related existing surveys

Review	Fuzzy / Evolutionary	Swarm Intelligence	PRISMA	SC Taxonomy
Minaee et al. (2023)	No	No	No	No
Pahuja & Goel (2024)	Partial	Partial	No	No
Singh et al. (2019)	No	No	No	No
Houssein et al. (2021)	No	No	Yes	No
Present review	**Yes (full)**	**Yes (full)**	**Yes**	**Yes**

2 Review Methodology

The present review was conducted using a structured and transparent methodology guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) framework [61]. The objective was to identify, screen, and synthesize research on soft computing techniques applied to biometric personal authentication, covering the period from 2010 to 2024.

2.1 Article Selection and Literature Search

A comprehensive literature search was performed using two major academic databases: **Scopus** (2010–2024) and the **Web of Science (WoS)** (2014–2024). These databases were selected for their broad coverage of peer-reviewed journals and conference proceedings in computer science, engineering, and applied mathematics and for their advanced query-filtering capabilities. The search combined keywords related to biometric authentication with terms describing soft computing paradigms using Boolean operators. Results were further refined through sequential filters applied to publication year, document type, and language.

2.2 Search Strategy and Query Construction

2.2.1 Scopus Search

The following query was executed in the Scopus title and abstract (TITLE-ABS) fields:

> TITLE-ABS (("biometric authentication" OR "multi-biometric authentication" OR "multimodal authentication") AND ("machine learning" OR "soft computing"))

This query returned 436 records. A publication-year filter was then applied to retain only papers issued between 2010 and 2024, reducing the count to 281 records. Finally, records were restricted to journal articles and conference papers published in English, excluding data papers, editorials, and retractions, yielding a final Scopus pool of **264 records**.

2.2.2 Web of Science Search

The following topic-field (TS) query was executed in WoS:

> TS ((“biometric authentication” OR “multi-biometric authentication” OR “multimodal authentication”) AND (“machine learning” OR “soft computing”))

This query returned 370 records. Applying a year filter restricted to 2014–2024 reduced the count to 339 records. Subsequent filtering by document type (retaining only journal articles and conference papers, and excluding early access, data papers, editorial material, and retracted publications) reduced the count to 329 records. A final language filter limiting results to English-language publications yielded a final WoS pool of **327 records**.

2.3 PRISMA Screening Procedure

The article selection process adheres to the PRISMA 2020 guidelines [61], and proceeds through four sequential phases: Identification, Screening, Eligibility, and Inclusion, as illustrated in Figure 4. The 264 Scopus records and 327 WoS records were merged and deduplicated, yielding a combined pool of **591 unique records**. Of these, 227 were excluded at the title and abstract screening stage as clearly out of scope—for example, records focused exclusively on forensic biometrics, medical imaging diagnostics, or general-purpose machine learning without a biometric authentication context. The remaining **364 records** were retrieved in full text for eligibility assessment. Full-text review led to the exclusion of a further 211 records for the following reasons: out of scope or not focused on biometric authentication as the primary application ($n = 103$); insufficient methodology, absence of reproducible experimental results, or conference extended abstracts ($n = 63$); and duplicated findings between the two databases that had survived automated deduplication ($n = 45$). The final review corpus therefore comprises **153 studies**.

2.4 Inclusion and Exclusion Criteria

Studies were included if they were published in a peer-reviewed journal or indexed conference proceeding, focused primarily on soft computing techniques—encompassing machine learning, deep learning, fuzzy logic, evolutionary algorithms, or swarm intelligence—applied to biometric authentication, and reported at least one quantitative performance metric such as accuracy, EER, FAR/FRR, or GAR. The search was restricted to papers published between 2010 and 2024 and available in full text in English. Biometric modalities considered include face, fingerprint, iris, palmprint, voice, gait, ECG, EEG, and multimodal combinations thereof. Studies were excluded if they were review papers, editorials, or opinion pieces without original experimental results; focused exclusively on non-authentication biometric tasks such as forensics or medical diagnosis; published outside the 2010–2024 window; unavailable in English full text; lacked reproducible experimental results; or represented duplicated findings already captured from the other database. Inclusion decisions were made independently by two reviewers, with disagreements resolved through discussion and, where necessary, referral to a third reviewer.

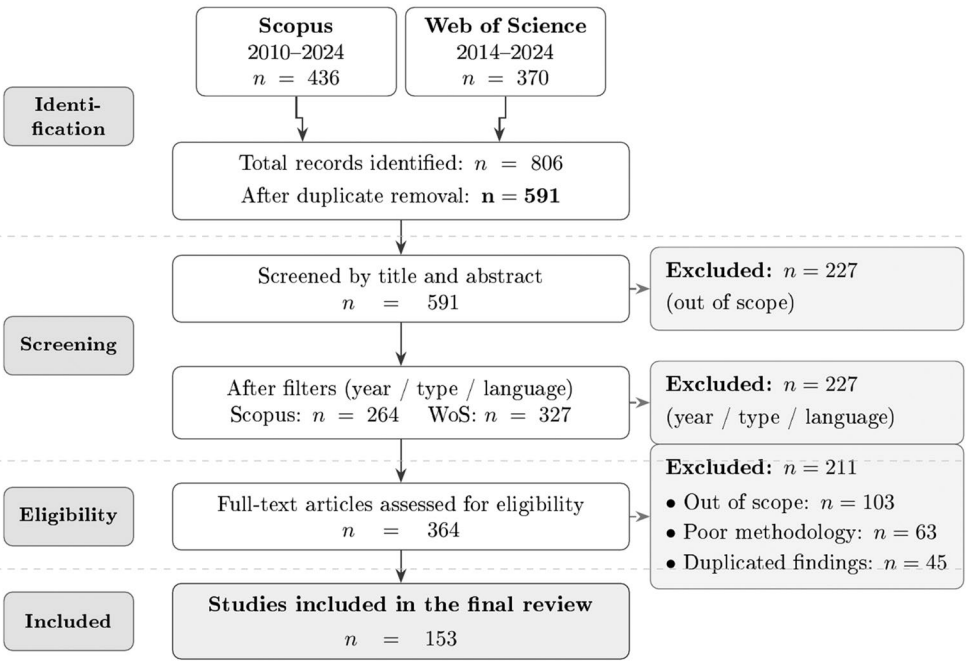


Fig. 4 PRISMA 2020 flow diagram illustrating the four-phase literature screening process. WoS = Web of Science

2.5 Analysis of Search Results

The temporal distribution of the 153 included studies reflects the growing research interest in applying soft computing methods to biometric data processing. As shown in Figs. 5 and 6, the volume of relevant publications accelerated substantially from 2018 onwards, coinciding with the widespread availability of large-scale biometric datasets and the maturation of deep learning frameworks. Swarm optimization algorithms and machine learning techniques account for the largest proportion of included studies, followed by fuzzy logic-based approaches and hybrid architectures. Computer science, engineering, and mathematics collectively represent the dominant subject areas, consistent with the technical focus of soft computing in biometric authentication. In recent years, increasing attention has been devoted to using soft computing methods for biometrical data processing, like swarm optimization algorithms (SOA) and machine learning. In addition, several machine learning methods and metaheuristic algorithms have been introduced in the past ten years for the analysis of biometrical data. This study explores the latest advancements in deep learning architectures, prominent metaheuristic algorithms, and machine learning techniques that demonstrate significant potential for applications in biometric systems.

2.6 Research Questions

This study addresses seven research questions designed to examine distinct aspects of soft computing in biometric authentication, focusing on comparative evaluation, methodological insights, and future research priorities.

- **RQ1** asks which soft computing techniques demonstrate the most favourable accuracy–efficiency trade-off in biometric feature extraction and classification across different modalities.
- **RQ2** examines how feature-level, score-level, rank-level, and decision-level fusion strategies compare in recognition accuracy and computational overhead when integrated with soft computing methods.
- **RQ3** considers which benchmark databases are most frequently used in biometric authentication research, and what their key limitations are regarding demographic diversity, environmental variability, and ethical data handling.
- **RQ4** investigates how meta-heuristic and swarm intelligence algorithms contribute to feature selection optimization in biometric systems, and what measurable performance gains they yield over baseline methods.

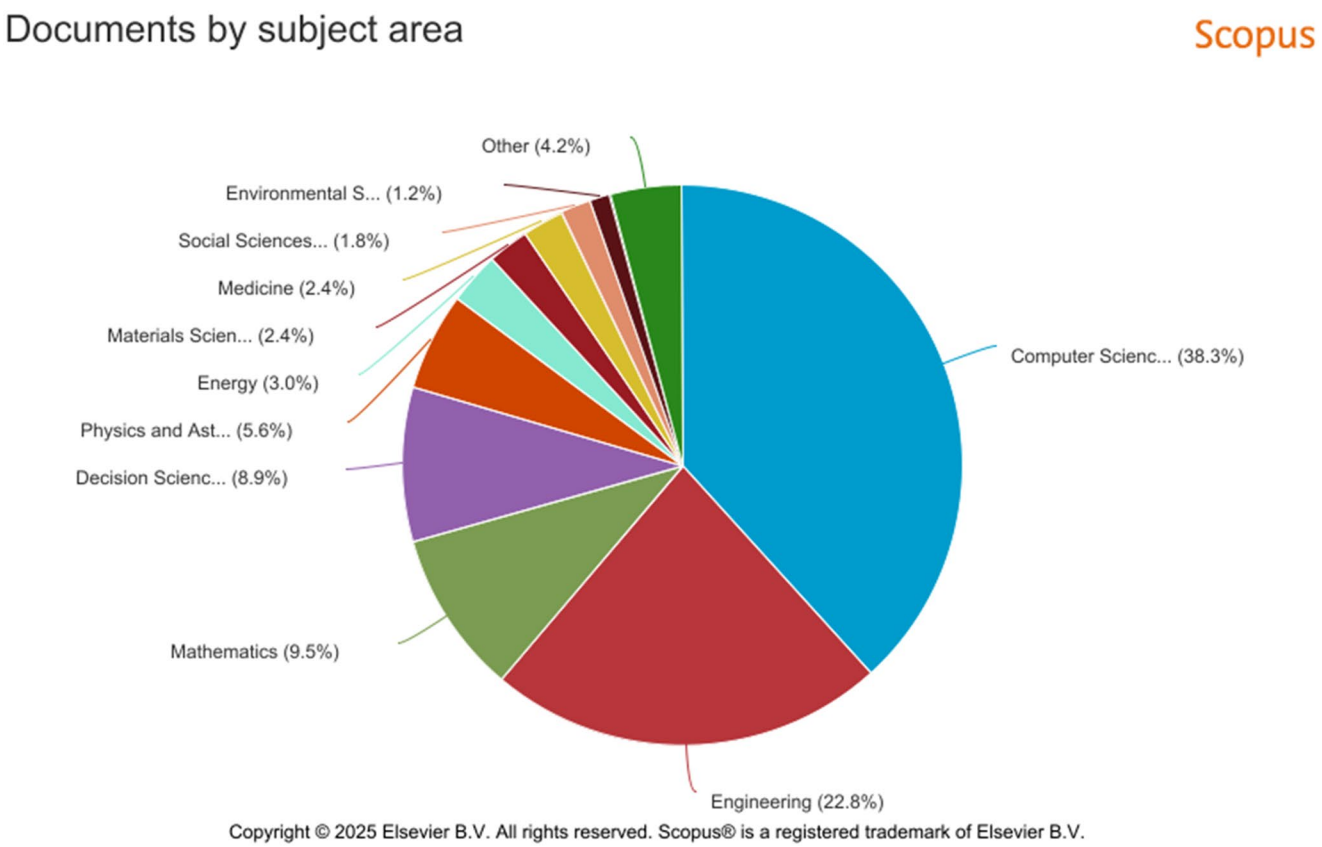


Fig. 5 Several graphical studies for machine learning in the field of biometrics

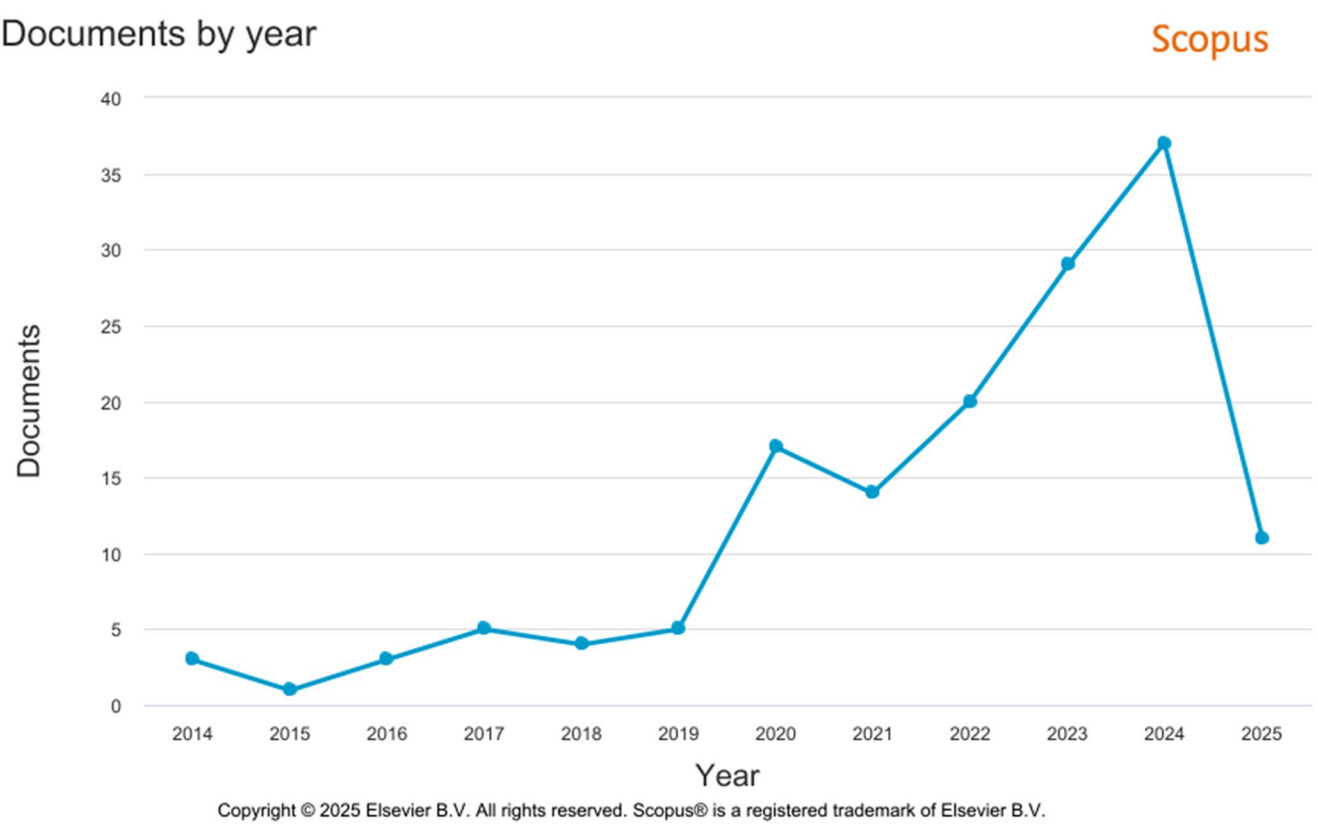


Fig. 6 Histogram of machine learning for biometric classification

- **RQ5** asks what deep learning architectures are most effective for multimodal biometric classification, and how their performance metrics compare against classical machine learning approaches.
- **RQ6** examines which hybrid soft computing frameworks—integrating neural networks, fuzzy logic, and evolutionary or swarm methods—achieve superior recognition performance, and what design principles enable such performance.
- **RQ7** considers what evaluation metrics are standardly employed in biometric authentication research, and how they collectively reflect the security–usability trade-off inherent in system design.

By addressing these questions, this review aims to provide a comprehensive and analytically grounded synthesis of the current state of soft computing for biometric authentication.

3 Fundamentals and Background

In this section, we explore the central focus of the investigation and present a detailed discussion of various concepts and methodologies commonly employed in the analysis of biometric systems.

3.1 Soft Computing Techniques

Soft computing encompasses a broad range of computational techniques, including artificial intelligence (AI), SOA, machine learning (ML), evolutionary computation, deep learning, fuzzy logic, and artificial neural networks (ANNs). In some fields, they are regarded as the most extensively used techniques. These techniques look for the most advantageous option [62]. This review will go further into specifics about the ML and SOA.

3.2 Machine Learning Techniques

Machine learning has become a core component of modern biometric authentication systems, enabling the automatic extraction of discriminative patterns from physiological and behavioral biometric traits. Its applications span feature extraction, classification, identity verification, multimodal fusion, and adaptive decision making across diverse biometric modalities [58–60].

Machine learning is typically categorized into supervised, unsupervised, and semi-supervised paradigms. In biometric systems, each paradigm addresses a distinct operational constraint. Unsupervised pre-training is particularly valuable because labeled biometric databases are typically

small—a consequence of the cost and privacy constraints associated with biometric data collection. By learning latent representations from unlabeled samples, unsupervised methods enable downstream classifiers to generalize from limited labeled enrolment data [63]. Semi-supervised learning is especially suited to cross-session biometric verification, where only enrolment (gallery) samples carry identity labels while probe samples remain unlabeled during testing; self-training and co-training models exploit this structure to progressively refine decision boundaries without costly re-annotation [64]. Supervised methods dominate the final classification stage, where labeled identity data is available [60].

Unsupervised Learning: This approach employs machine learning methods to examine and group unlabeled datasets. These algorithms autonomously detect hidden patterns or data clusters without human intervention [65].

In biometric authentication, unsupervised learning is commonly used for clustering, representation learning, and dimensionality reduction when labeled biometric samples are limited or unavailable. By discovering latent structures in biometric data, these methods can improve feature extraction and support the development of more robust authentication models [64]. The two principal categories of unsupervised learning considered in biometric applications are clustering and association analysis.

Numerous algorithms have been proposed to address these challenges; nevertheless, popular strategies are predicated on: Expectation-maximization algorithm, k-means, Gaussian Mixture Models, Hebbian Learning approaches, Convolutional Neural Networks (CNNs).

Unsupervised algorithms for biometric applications primarily concentrate on protecting individual biometric data through encryption [66, 67], feature-level fusion [68], extraction of biometric data meaning, and behavioral pattern recognition, among other methods. Furthermore, biometric systems that employ unsupervised learning to precisely localize biometric features guarantee improved registration and learning policy construction, which in turn permits improved categorization. A notable example of a successful speaker verification system developed at MIT Lincoln Laboratory utilized a Universal Background Model (UBM) consisting of 2048 Gaussian components with diagonal covariance matrices [49, 69]. Furthermore, an iterative Expectation-Maximization (EM) algorithm was integrated into the Tardos fingerprint recognition code to adapt collusion strategies. In addition, Vlachos and Dermatas [70] introduced the Nearest Neighbor Clustering Algorithm (NNCA), an innovative unsupervised clustering method that has been effectively applied to retinal vessel segmentation. It can be utilized for the fully automatic extraction of finger vein patterns because it operates autonomously.

Clustering: To make the members of each group more similar to one another and distinct from those in other groups, the population or set of data points is clustered. It is just a collection of objects based on their similarities and contrasts [71]. Clustering algorithms such as K-means and K-median are commonly employed in the field of biometrics for tasks such as pattern recognition and data segmentation.

Association Rules: Association rules provide a rule-based mechanism for discovering meaningful relationships among variables within biometric datasets. In biometric applications, association rule mining can be used to identify correlations among extracted features, support multimodal fusion strategies, and reveal behavioral patterns that improve authentication performance. AIS, SETM, Apriori, and their variants are among the most widely used association rule mining algorithms [72].

Table 2 presents a summary of selected recent studies that employ unsupervised learning techniques in biometric systems, along with the corresponding outcomes of each approach. In conclusion, unsupervised learning plays a complementary role in biometric authentication by supporting feature discovery, dimensionality reduction, clustering, and representation learning. Although it rarely performs final identity verification directly, it can significantly improve the effectiveness of downstream classification and fusion stages.

Reinforcement Learning: Although reinforcement learning is less frequently adopted than supervised and unsupervised approaches in biometric authentication, it has attracted increasing attention for adaptive decision-making, feature selection, dynamic threshold optimization, and continuous system learning. By leveraging reward-based feedback mechanisms, reinforcement learning enables biometric systems to adapt to changing user behavior and operational conditions without requiring complete retraining [80, 81].

In biometric systems, reinforcement learning has been investigated for classification tasks [80], continuous training using reward and punishment feedback signals [81], identifying discriminative biometric characteristics [82], and feature extraction. Reinforcement learning offers flexibility in scenarios where biometric systems must continuously update their decision strategies based on new observations and environmental changes [83, 84].

However, traditional reinforcement learning methods often face difficulties when applied to high-dimensional biometric data. Recent advances in Deep Reinforcement Learning (DRL) have partially addressed this limitation by combining reinforcement learning with deep neural networks. Despite these advances, reinforcement learning remains relatively underexplored in biometric authentication compared with supervised and deep learning approaches,

Table 2 Biometrics applications of unsupervised learning techniques

Ref	Description	Biometric Application	Algorithm	Results
[73]	Proposes a Distributed Speaker Voice Activity Detection (DM-VAD) approach for Wireless Acoustic Sensor Networks (WASNs), using energy signal unmixing to localise sound sources.	Speaker voice activity detection	K-Means, K-Medoids, and K-Medians clustering	VAD accuracy of 85% under challenging conditions (20 nodes, 7 sources, reverberant room)
[74]	Proposes an unsupervised method to estimate a unique rotational offset for each fingerprint sample using CNNs trained without orientation labels.	Fingerprint alignment and recognition	Unsupervised CNN for rotational offset estimation; multiple live-scan datasets	Mean absolute deviation of 2.1° on matched training data; 6.2° on unseen datasets
[75]	Proposes ArcFace, an additive angular margin loss for discriminative deep face representation learned without manual label supervision of the margin space.	Face recognition	ArcFace loss + deep CNN; LFW, YTF, MegaFace datasets	99.83% accuracy on LFW; 98.02% on YTF
[76]	Presents GaitPrivacyON, a mobile gait biometric verification framework that preserves subject privacy while maintaining authentication performance, using convolutional autoencoders for unsupervised latent representation learning.	Gait-based biometric authentication	Dual convolutional autoencoders (shared weights) + Siamese CNN–RNN for gait verification; OU-ISIR, MotionSense, MobiAct datasets	AUC $>$ 96.6% across all datasets with strong privacy preservation
[77]	Proposes a computer vision-based individual identification system using only the victory hand gesture (two fingers), targeting terrorist identification scenarios.	Hand gesture biometric identification	Otsu's method, K-means (colour-based), and ANN for unsupervised hand segmentation	100% hand segmentation accuracy; 93% overall identification accuracy
[63]	Proposes an unsupervised DBM approach for dorsal hand vein (DHV) recognition, with CLAHE and median filtering preprocessing, followed by CNN-based feature extraction without labeled data.	Hand vein recognition (DHV)	Unsupervised Deep Boltzmann Machine (DBM) + CNN; two public DHV datasets	Accuracy 99.9%, recall 99%, precision 99%, F-score 99%
[78]	Proposes PerAE, a personalized autoencoder for ECG-based biometric authentication in AR/VR environments. Each user's model learns reconstruction of their own signal; other users' signals are flagged as anomalies via reconstruction error.	ECG-based biometric authentication	Personalized Attention-based Memory Autoencoder (Attention-MemAE); unsupervised anomaly detection	90% identification accuracy using $\sim$500 heartbeat samples ($\approx$ 5 min) per user
[79]	Employs a CNN autoencoder in a fully unsupervised manner to compress and reconstruct fingerprint images, enabling discriminative feature learning without labels. Evaluated across four fingerprint datasets for cross-sensor generalization.	Fingerprint recognition	CNN Autoencoder (unsupervised reconstruction); four fingerprint datasets	Accurate reconstruction across all four datasets; superior handling of overlapping and low-quality ridge patterns

highlighting an important opportunity for future research in adaptive and resource-aware biometric systems [85].

Supervised learning: In supervised learning, algorithms are trained using labeled datasets to enable accurate prediction of outcomes or correct data classification. In biometric authentication, supervised learning is commonly used to classify biometric samples into known identities or to verify claimed identities using labeled enrolment data. With this method, the models are trained and guided toward generating the intended results using a training set that consists of inputs and matching labeled outputs [86]. A loss function is used to gauge the algorithm's correctness, and iterations are carried out to lower the error until it is within an acceptable bound. Regression and classification are the two additional categories into which supervised learning falls.

Supervised learning, in contrast to unsupervised learning, is primarily employed during the final stages of a biometric recognition system. Supervised learning is concentrated on classification and regression, whereas unsupervised techniques are utilized for finding clusters, latent components, graph structure, and matrix completion. For biometric modalities fusion [87], biometric data categorization [88], and regression for dependable [89], effective, and safe multibiometric systems [90], supervised learning has been demonstrated to be helpful. Supervised learning remains the dominant paradigm in biometric authentication because authentication decisions ultimately require identity classification or verification. Consequently, supervised models such as SVM, k-NN, decision trees, and deep neural networks are widely employed to learn discriminative representations from labeled biometric data and to support robust recognition across different biometric modalities [60, 90]. Recent reviews on multimodal biometric fusion further emphasize that supervised learning models remain central to integrating heterogeneous biometric modalities because they can learn discriminative relationships among

fused feature representations and matching scores. This capability is particularly important in multimodal systems, where information from different biometric traits must be combined to improve recognition accuracy, robustness, and resistance to noisy or incomplete biometric samples [91].

Support Vector Machines (SVMs): SVMs are a prominent supervised learning approach frequently applied in classification tasks due to their high accuracy [92]. They achieve this by utilizing kernel functions to map input data into a higher-dimensional feature space, enabling the formation of nonlinear decision boundaries that effectively separate distinct classes. The main goal of SVMs is to determine an optimal hyperplane that efficiently distinguishes data points into their respective categories. This goal is realized by enlarging the margin between the separating hyperplane and the nearest data instances from each class, identified as support vectors. In a two-dimensional space, the hyperplane acts as a decision boundary that divides the plane into two distinct regions, each corresponding to a different class. The performance and flexibility of an SVM model are influenced by several hyperparameters, including the regularization parameter C and the kernel coefficient γ, which are defined during the construction of the classifier. These tuning parameters, carefully selected to prevent underfitting or overfitting, affect the SVM's performance. It is important to note, however, that standard SVM formulations incur $O(n^2)$ to $O(n^3)$ training complexity with respect to the number of training samples, making them poorly scalable to large-scale multimodal biometric databases. This scalability constraint motivates the use of approximate kernel methods or swarm-optimized feature selection to reduce dimensionality prior to SVM training [59].

K-nearest neighbors (K-NN): k-NN approach can be used for both regression and classification in applications that need prediction [93]. Applications for classification problems are common. It has several advantages, such as quick computation speeds, simple output interpretation, and forecasting capability. The k-NN method is a supervised learning technique used to classify unknown samples based on their similarity to k nearest neighbors. It is also a prominent machine learning approach for pattern recognition and interpretation. It provides a classification based on the acceptable relevance of the chosen features. In biometric authentication, KNN remains attractive because of its simplicity and effectiveness in multimodal score classification, although its computational cost increases significantly with database size. Consequently, KNN is more suitable for small- to medium-scale biometric deployments than for large national-scale identity systems. In the k-NN algorithm, the parameter k specifies how many nearest neighbors are taken into account to classify a given data point. The algorithm uses a distance metric—commonly Euclidean distance—to evaluate the proximity between points based on their feature values. Choosing a suitable value for k helps to avoid issues with overfitting and underfitting. Unlike models that explicitly learn a decision boundary, the k-nearest neighbors (k-NN) classification algorithm makes predictions based on the closest training instances in the feature space [94].

In [95], an effective multimodal biometric system is proposed, utilizing the concatenation of matching scores of left palmprint, right palmprint, and face modalities. The system employs CNNs for feature extraction and k-Nearest Neighbors (k-NN) for classification. This multimodal approach is designed to enhance person identification and recognition by integrating biometric scores from multiple sources. The FEI face dataset and the IITD palm print database, two well-known biometrics benchmarks, are utilized as raw materials to train biometric algorithms to create robust and secure verification/identification systems. To assess the resilience of the proposed model in adverse or challenging scenarios, experiments are conducted using noisy biometric datasets. The CNN and KNN multi-modal biometric systems surpass the majority of the most widely used contemporary biometric verification approaches, according to computer simulation data.

With the use of both fingerprint and iris scans, the study in [96] advances the integration and implementation of machine learning methods for biometric recognition for wireless sensor networks (WSN). As briefly discussed in this study, machine learning algorithms are employed to enhance recognition accuracy following the extraction and feature-level fusion of fingerprint and iris characteristics. Given that the system is primarily intended for Wireless Sensor Networks (WSNs), feature-level fusion is essential to reduce computational overhead and improve processing efficiency. Over 95% accuracy was attained by the suggested strategy using all machine learning algorithms.

Naive Bayes Classifier: The Bayes theorem forms the foundation of the naïve Bayes classifier. Given the probabilistic classifier backgrounds of the attributes, it provides a significant degree of independence between them. Due to their ease of implementation, these classifiers are frequently employed in machine learning categorization. The Eq. (1) is used to construct this classifier.

$$P(A \mid B) = \frac{P(B \mid A)(P(A)}{P(B)} \tag{1}$$

The Gaussian model, a type of Naive Bayes classifier, assumes that continuous data follow a normal distribution. This model simplifies the process of deriving statistical outcomes from the training dataset [97].

To address the limitations of conventional PPG feature extraction, Yang et al. [98] proposed a biometric recognition

framework combining multi-feature extraction with a Naive Bayes classifier. Experiments conducted on three PPG datasets reported recognition rates exceeding 97%, demonstrating the suitability of Naive Bayes for lightweight physiological biometric applications.

In biometric authentication, Naive Bayes is particularly attractive for wearable and physiological biometrics such as PPG and ECG because it offers low computational complexity and fast inference. However, its conditional independence assumption may limit performance when strong correlations exist among extracted biometric features.

Regression methods: Regression techniques rely on a statistical framework that determines the extent of correlation between variables. Regression analysis is a valuable method for concluding data and has also been applied to forecast future results. This approach is used in many different fields. The regression approach has three methods: ridge regression, ordinary least squares, and linear regression [99].

Linear regression, a core and extensively utilized predictive modeling technique, seeks to establish a linear association between one or multiple variables. Using the value of one variable as a guide, one can forecast the value of another. There are two types of linear regression: simple regression and multiple regression (MLR) [100]. In biometric authentication, logistic regression is frequently used for classification and score-level fusion because of its simplicity, interpretability, and computational efficiency. Nevertheless, its capacity to model complex nonlinear relationships among heterogeneous biometric features remains limited, particularly in multimodal authentication scenarios where deep learning architectures have demonstrated superior representational capabilities [59, 101].

Table 3 presents a comprehensivetermine the most effective solution [107]. To improve the efficiency of search procedures, SOAs use a niques in biometric systems across various modalities. The studies span from thermal-visible face recognition and ECG-based authentication to multimodal systems involving fingerprint, iris, and facial features. Deep learning models, particularly convolutional neural networks (CNNs), are prominently utilized due to their strong discriminative capabilities in image and signal processing tasks. For instance [102], successfully narrows the modality gap in thermal-visible face recognition using a deep neural approach, achieving significant improvements over prior methods. Likewise [40, 103], demonstrate the robustness of temporal and recurrent deep learning architectures for ECG-based identification and verification. Additionally, hybrid systems such as the ESVM-KM technique [104] highlight the effectiveness of ensemble support vector machines in enhancing multimodal biometric accuracy. These findings underscore the versatility and high accuracy of supervised learning methods in advancing biometric authentication and identification systems.

A notable example of CNN-based supervised classification in a clinical biometric context is the work of [108], which proposes a shallow CNN architecture trained on ECG

Table 3 Biometrics applications of supervised learning techniques

Ref	Description	Biometric application	Algorithm	Results
[102]	Proposes a deep neural network to model the non-linear correlation between thermal and visible face modalities, reducing the cross-modal gap for face recognition.	Thermal–visible face recognition	Deep perceptual mapping network; UND-X1 and NVESD datasets	Rank-1 accuracy improvement of >10% on UND-X1 and 15–30% on NVESD; modality gap reduced by >40%
[103]	Proposes TCNN and RNN architectures for ECG-based biometric identification and authentication.	ECG biometric identification and authentication	Temporal CNN (TCNN) and RNN; multiple ECG datasets	TCNN achieves ≈100, 96, and 90% identification accuracy; EER of 0.0%, 0.1%, and 2.2% for authentication
[40]	Builds a multi-scenario ECG biometric database covering different days, activities, and positions. Deep learning is used for classification without signal manipulation.	ECG biometric verification (BioECG)	Deep Learning network; multi-day, multi-activity ECG dataset	Two-day enrolment improved performance; EER increased 63% on posture change, 99% on cross-day, 91% post-exercise
[104]	Proposes ESVM-KM, an ensemble SVM with Gabor wavelet feature extraction for multimodal biometric recognition using fingerprint, face, and iris.	Multimodal biometric recognition (fingerprint, face, iris)	Ensemble SVM with kernel mapping (ESVM-KM) + Gabor wavelet features	Higher recognition rates and lower computation time versus state-of-the-art methods
[105]	Applies EMD for ECG denoising and ROI extraction, followed by multi-domain feature extraction (statistical, time, and frequency). Multiple classifiers are compared.	ECG biometric authentication	SVM (cubic kernel), k-NN, Decision Tree; 14-subject dataset (BIOPAC)	Best: SVM cubic kernel—accuracy 98.7%, sensitivity 100%, specificity 98.8% (10-fold CV)
[106]	Proposes a deep palmprint recognition method using self-supervised contrastive pretraining followed by supervised fine-tuning with an uncertainty-aware loss, addressing label scarcity in unconstrained conditions.	Palmprint recognition	Self-supervised contrastive pretraining + uncertainty loss; fine-tuned CNN; standard palmprint benchmarks	Significant accuracy improvement under unconstrained conditions; outperforms supervised baselines

signals from leads II, III, and AVF to differentiate healthy cardiac activity from inferior myocardial infarction (IMI). Evaluated on the PTB database using a subject-specific protocol, the model achieved an accuracy of 84.54%, sensitivity of 85.33%, and specificity of 84.09%, outperforming existing benchmark approaches. Although primarily a clinical detection task rather than personal authentication, this work demonstrates the transferability of supervised deep learning architectures to physiological signal-based biometric analysis.

3.3 The Swarm Optimization Algorithms

Swarm intelligence (SI) algorithms address optimization challenges in biometric systems that classical gradient-based methods cannot efficiently solve, particularly in high-dimensional feature selection, multi-modal fusion weight assignment, and classifier hyperparameter tuning [109]. Unlike deterministic optimization techniques, SI algorithms explore large search spaces without becoming trapped in local optima, making them well-suited to the non-convex landscapes typical of biometric feature spaces [110].

Three biometric-specific contributions of swarm intelligence methods are particularly well-evidenced in the reviewed literature. First, in **feature subset selection**, PSO-based methods achieve accuracy gains of 3–8% points over full-feature baselines across fingerprint, iris, and EEG modalities [109, 111], by eliminating correlated and low-discriminative features that degrade classifier performance in high-dimensional biometric spaces. Second, for **score-level fusion weight optimisation**, hybrid PSO–ACO methods consistently outperform uniform weighting strategies in multimodal systems combining face, fingerprint, and iris matchers [112]. Third, for **CNN and SVM hyperparameter tuning**, swarm methods reduce manual grid-search cost while achieving competitive or superior accuracy – the Bat Algorithm-optimized Deep Belief Network of Basha et al. (2020) achieved higher recognition rates than default-parameterised baselines on a three-modality benchmark [113].

In biometric applications, SI methods operate primarily as optimization layers integrated with classifiers rather than standalone recognition systems. Their three principal contributions are: (1) feature subset selection to reduce dimensionality while preserving discriminative information; (2) fusion weight optimization at the score or decision level; and (3) hyperparameter tuning for SVM, ANN, and CNN classifiers [59, 109].

3.4 Biometric and MultiBiometric Datasets Analysis

Reliable biometric authentication systems depend heavily on access to diverse and high-quality datasets. These datasets are fundamental for training, validating, and benchmarking algorithms across various biometric modalities. Key aspects such as the number of subjects, variety in acquisition conditions, and intra-class variability significantly impact a system's performance and robustness.

3.4.1 Face Recognition Datasets

Facial recognition rSwarm intelligence techniques, inspiredemains a prominent area in biometrics due to its nonintrusive nature and ease of data acquisition. Several widely used datasets in this domain include

- **Labeled Faces in the Wild (LFW)**: Featuring over 13,000 face images collected in real-world conditions, LFW is used to evaluate algorithms under uncontrolled settings [114].
- **CASIA-WebFace**: A large-scale dataset with nearly half a million images of over 10,000 individuals, it has been instrumental in advancing deep learning models for facial recognition [115].
- **VGGFace2**: This dataset provides more than 3 million images from over 9,000 identities, covering a broad range of pose, age, and ethnic diversity [116].

3.4.2 Fingerprint Datasets

Fingerprint recognition systems are widely deployed, especially in forensic and security applications. Common benchmark datasets include:

- **FVC2002/FVC2004**: These datasets, compiled during international fingerprint verification competitions, include samples from different sensors and synthetic images to test system versatility [117].
- **NIST SD4 and SD14**: Developed by NIST, these datasets provide high-resolution rolled fingerprints frequently used in law enforcement systems [118].

3.4.3 Iris Recognition Datasets

Due to the unique and stable nature of iris patterns, iris recognition is considered highly reliable. Datasets in this area include:

- **CASIA-IrisV3**: Offers images captured under varied environmental conditions, supporting robustness studies [119].

– **UBIRIS.v2**: Focuses on less constrained iris recognition with visible light images that include noise factors like blur and reflections [120].

3.4.4 Voice and Gait Datasets

Biometric traits such as voice and gait are behavioral and can be acquired passively. Notable datasets include:

– **TIMIT**: A well-known speech corpus containing phonetically diverse audio samples, often used in speaker recognition research [121].
– **CASIA Gait Dataset**: Contains video sequences of individuals walking under different conditions, enabling gait recognition studies [122].

3.4.5 Multimodal Biometric Datasets

Multimodal datasets provide data from multiple biometric sources for the same subject, supporting fusion-based approaches:

– **MULB**: Contains synchronized samples of fof face, iris, and hand geometry from a diverse set of subjects [123].
– **WVU Multimodal Dataset**: Collects face, voice, fingerprint, iris, and palmprint data in a real-world setting, facilitating cross-modal studies [124].
– **BioSecure DS2**: A European dataset covering face, voice, and fingerprint samples from over 700 participants [125].

3.4.6 Challenges and Ethical Considerations

Although existing datasets have propelled research forward, they still face notable limitations. These include limited demographic diversity, small sample sizes, and controlled capture settings that do not reflect real-world complexity. Additionally, concerns surrounding the privacy, consent, and ethical handling of biometric data are increasing. Recent research trends focus on privacy-preserving machine learning, synthetic dataset generation, and decentralized learning approaches to mitigate these issues.

3.5 Evaluation Metrics

Performance evaluation in biometric authentication systems relies on two complementary sets of metrics: general classification metrics and biometric-specific error rates.

$$\text{Accuracy} = \frac{TV + TR}{TV + TR + FV + FR} \tag{2}$$

where TV, TR, FV, and FR denote the number of true positives, true negatives, false positives, and false negatives, respectively.

$$\text{Sensitivity} = \frac{TV}{TV + FR} \tag{3}$$

where TV is the number of correctly identified genuine users and FR is the number of genuine users incorrectly rejected.

$$\text{Precision} = \frac{TV}{TV + FV} \tag{4}$$

where FV denotes the number of impostors incorrectly classified as genuine users.

$$F\text{-score} = \frac{2 \times \text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{5}$$

The F-score provides a single measure that balances both Precision and Sensitivity, and is particularly useful when the dataset is imbalanced.

3.5.1 Performance Estimation of Biometric Authentication Systems

The Hamming distance serves as a metric to assess the similarity between binary feature vectors originating from the stored database and biometric query templates.

For feature vectors extracted from images of the same person, the Hamming distance between two samples is generally close to zero. Conversely, for an impostor user, the feature vectors are typically random and exhibit a Hamming distance of approximately 0.5 when they originate from different individuals. False Accept Rate (FAR), False Reject Rate (FRR), and Equal Error Rate (EER) are common performance indicators that can be used to evaluate the accuracy of the biometric recognition infrastructure.

– The False Rejection Rate (FRR), also known as the False Non-Match Rate (FNMR), represents the percentage of instances in which an authentication system incorrectly rejects a legitimate user as an impostor, which is determined by Eq. 6

$$FRR = \frac{n_R}{N} * 100 \tag{6}$$

where N represents the total number of tested users, and n_R denotes the number of users incorrectly rejected.

Table 4 Biometric authentication evaluation metrics: definitions and security–usability implications

Metric	Formula	Security / Usability	Typical / Range
FAR	$n_{FA}/N_I \times 100$	Low = high security; raises FRR	0.001–2%
FRR	$n_R/N \times 100$	Low = high usability; raises FAR	0.1–5%
EER	FAR = FRR	Lower = better balance	0.1–5%
GAR	100 − FRR	Higher = better usability	95–99.9%

– The False Acceptance Rate (FAR), also known as the False Match Rate (FMR), quantifies the percentage of instances in which an authentication system erroneously accepts an impostor as a legitimate user. This value is computed as shown in Eq. (7)

$$FAR = \frac{n_{FA}}{N_I} * 100 \tag{7}$$

where N_I represents the total number of tested impostors, and n_{FA} denotes the number of impostors incorrectly accepted.

– Genuine accept rate (GAR): This is established by calculating the proportion of actual registered users that the authentication infrastructure recognizes. Eq 8 can be used to calculate this.

$$\begin{aligned} GAR1 &= \frac{n_{GA}}{N_{GT}} \\ GAR &= 100 - FRR \end{aligned} \tag{8}$$

where n_{GA} is the total number of accepted genuine users, and N_{GT} is the total number of genuine trials.

– Equal Error Rate (EER): This metric reflects the trade-off between the rates of false rejections and false acceptances, illustrating their balance within the authentication system. A lower EER value indicates better overall system performance. The EER also known as the Crossover Error Rate or Crossover Rate, refers to the point where the false acceptance rate and false rejection rate are equal in an authentication system. The following is how the Eq. 9 calculates the EER:

$$EER = FAR, \, where \, FAR = FRR \tag{9}$$

A lower EER indicates better overall system performance. Table 4 summarizes these metrics alongside their typical ranges reported in the biometric authentication literature.

As the False Acceptance Rate (FAR) decreases, the False Rejection Rate (FRR) typically increases, and vice versa. The Equal Error Rate (EER) represents the point where the percentages of false rejections and false acceptances are identical. Attempting to minimize the FAR to its lowest feasible level may significantly elevate the FRR, indicating that heightened security through stricter access control can reduce system usability due to erroneous user rejections. Conversely, lowering the FRR to enhance user acceptance may compromise the authentication system's security. This trade-off is illustrated in Fig. 7, which compares FRR and FAR across different system thresholds.

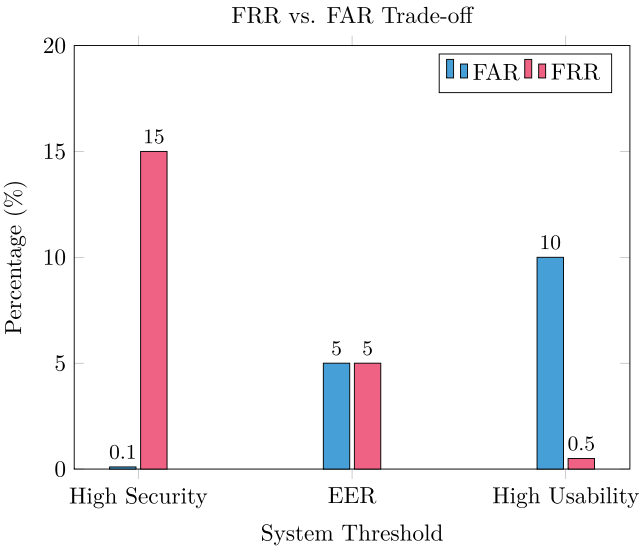


Fig. 7 Bar chart comparing FRR and FAR at different system thresholds

4 Soft Computing Methods for Multibiometric Applications

Soft computing methods that are frequently employed in the multibiometric sector are introduced in this section. Additionally, several multibiometric applications will be offered. We start by introducing the ML techniques based on multibiometric data analysis. Next, we introduce the swarm intelligence algorithms that are based on the analysis of multibiometric data.

4.1 Machine Learning and Deep Learning for multiBiometric Data Analysis

The study [50] introduces "SmartIris ML," a state-of-the-art method that improves multibiometric iris detection by utilizing machine learning. Their technology combines various modalities using an advanced machine learning framework, utilizing the unique and distinctive characteristics of the iris

to provide increased accuracy and dependability. This work examines the complexities of machine learning methods and their applicability to various iris biometric data sources. To create a more reliable and secure authentication system, they suggest a thorough fusion approach that best integrates the advantages of distinct iris modalities. SmartIris ML's machine learning algorithms respond to changing environmental conditions and demographic factors by learning from various datasets. The outcomes of studies on sizable datasets demonstrate SmartIris ML's efficacy in practical settings and point to its potential for use in identity verification, secure access management, and other fields needing strong authentication. In [126], the authors propose a face-ocular multimodal biometric system for gender prediction, utilizing both facial and ocular biometrics within a hybrid framework. The study explores the impact of multimodal biometric systems on gender classification at feature-level and score-level. To capture texture details from the face and ocular regions, the uniform local binary pattern (ULBP) feature extractor is employed. Additionally, a novel evolutionary algorithm, the backtracking search method, is introduced to select the most effective feature sets for both modalities. For classification, a support vector machine (SVM) integrates the combined facial and ocular features and scores. The proposed approach is validated using the CASIA-Iris-Distance and MBGC multimodal biometric databases, with a subject-disjoint training and testing protocol. The reported accuracy is 92.51% for MBGC and 88.00% for CASIA.

In [127], the spatial pixels (picture elements) inside the image are the subject of the recognition investigation. An attempt is made to recognize two distinct biometric features: face pictures and fingerprints. Every pixel in fingerprint images is examined. Additionally, Facial image analysis utilizes both the distances among distinct facial mark identities and the relational differences, including variations in distance and gonial angular measurements between the left and right mandible bones. This work employs principal component analysis, which goes beyond conventional methods, along with the curvelet transform. Good results are obtained using the multi-modal biometric technique with all security systems for facial recognition and fingerprint recognition.

4.2 Multibiometric Data Analysis Using Swarm Optimization Techniques

Swarm intelligence techniques, inspired by collective behaviors found in nature—such as bird flocking, insect swarming, and fish schooling—have emerged as valuable optimization tools in the domain of biometric systems. These algorithms are particularly effective in solving complex problems such as feature selection, classifier tuning, and fusion optimization within multibiometric environments.

Multibiometric systems leverage multiple biometric traits, such as face, fingerprint, iris, and voice, to achieve higher recognition accuracy, reliability, and robustness. However, the integration of diverse biometric modalities often introduces increased data dimensionality and computational challenges. Swarm-based algorithms are well-suited to these scenarios, as they offer efficient solutions for navigating large search spaces and optimizing performance metrics.

Particle Swarm Optimization (PSO) is one of the most widely adopted swarm techniques, modeled after the social behavior of birds. PSO has demonstrated excellent performance in selecting relevant subsets of characteristics and optimizing classifier parameters, particularly in systems that combine multiple biometric traits [128]. The authors of [129] introduced a biometric recognition framework that integrates various traits through the fusion of PSO-based features. Their method showed improved performance by selecting the most relevant features from different modalities, minimizing redundancy, and increasing the discriminative power of the combined biometric template.

Ant Colony Optimization (ACO), which simulates the foraging behavior of ants, is known for its success in the fusion of scores and decisions. Its iterative search based on pheromones allows an effective exploration of optimal fusion rules in multibiometric contexts [130].

More recently developed metaheuristics like the **Firefly Algorithm (FA)** and **Bat Algorithm (BA)** have gained attention for their speed, flexibility, and adaptability in high-dimensional optimization. These methods have shown promising results in multimodal biometric systems for tasks such as feature matching and classifier tuning [113, 131].

The study [132] suggested a technique for extracting the fingerprint and iris characteristics needed for a multimodal biometric authentication system. This system uses an SVM classifier for classification and a 1D Gabor filter for biometric attributes to extract the feature single modalities. The accuracy of the suggested approach is 93.4% when pattern matching for fingerprint and iris identification is done

Recent research also explores hybrid swarm algorithms, which combine the strengths of multiple optimization strategies. Hybrid approaches, such as PSO-ACO models, have improved fusion performance by optimizing weight assignment and rule structures for score-level integration [112]. Additionally, combinations with fuzzy logic or genetic algorithms offer more flexibility when handling uncertainty and nonlinearity in biometric data [133].

Swarm intelligence has also been successfully integrated into deep learning pipelines. Techniques such as swarm-optimized Deep Belief Networks (DBNs) and CNNs have enhanced feature selection and model training in complex biometric tasks. These hybrid systems benefit from the

global search efficiency of swarm algorithms and the hierarchical feature learning of deep networks, making them ideal for scalable deployment [113, 134].

Despite their strengths, swarm optimization methods face certain limitations. These include sensitivity to parameter settings, the risk of premature convergence, and occasional difficulties in escaping local optima. To address these issues, researchers are exploring adaptive parameter tuning, ensemble strategies, and hybrid metaheuristics. Continued innovation in this area is expected to significantly advance the reliability and performance of multibiometric authentication systems.

4.3 Applications-Based Soft Computing Techniques

Multibiometric authentication systems, which integrate multiple biometric modalities (e.g., face, fingerprint, iris, gait, voice), have gained prominence due to their robustness, accuracy, and resilience to spoofing attacks. Soft computing techniques, including neural networks, fuzzy logic, genetic algorithms, and deep learning, have further enhanced the adaptability and intelligence of these systems across diverse application domains.

4.4 Smartphone Authentication

Recent advancements in behavioral biometrics have contributed significantly to smartphone authentication. A 2025 study proposed a framework combining machine learning with the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) to analyse user interactions such as touch dynamics and motion patterns, yielding robust performance [7]. Another work utilized a hybrid deep learning model that continuously monitors sensor data (accelerometer, gyroscope) for real-time user authentication [135].

4.5 Surveillance and Security Systems

In surveillance scenarios, soft computing techniques have been used to fuse face and gait modalities. An attention-based deep neural network architecture was employed to adaptively integrate these modalities under varying environmental conditions, improving identification accuracy and system robustness [136].

4.6 Online Security and Access Control

A novel framework for multi-modal authentication that is based on many users is created and applied to huge image data types. Several biometric traits, including fingerprint, face, and IRIS features, are employed in this framework to determine a user's unique integrity for data security and authentication. To ensure robust data security and integrity verification in remote systems, A new computational integrity algorithm and encryption technique are employed. Experimental results reported in [137] show that the proposed multi-modal integrity-based encryption model achieves a computational integrity bit change exceeding 7% and operates with a runtime of only 5% on large datasets.

4.7 Financial and Governmental Applications

For high-security environments such as banking and e-governance, a Fine-Tuned Cuckoo Search Optimized Convolutional Neural Network (FCSO-CNN) was developed to fuse voice and iris biometrics. This multimodal system achieved high recognition accuracy, offering a scalable solution for secure identity verification [138].

The study in [139] proposes a hybrid voice authentication framework designed to enhance security against forgery attacks in mobile banking systems. The approach enables users to verbally input a numeric password, which is decoded from acoustic signals using an artificial neural network (ANN). Key speech features, such as energy levels and Mel Frequency Cepstral Coefficients (MFCCs), are extracted to aid in authentication. The system detects individual digits through pause-based segmentation and matches the input with stored credentials using frequency and phase characteristics to resist spoofing attempts. A fuzzy nonlinear SVM classifier, trained with the Ali Baba and the Forty Thieves algorithm, underpins the verification process. Evaluated across a dataset involving 30 users and three smartphones, the proposed system achieved an accuracy exceeding 98.29%. It demonstrates robustness against various real-world challenges, including changes in speaking angle, distance, passphrase length, background noise, and device type, indicating strong generalization and resilience.

4.8 Internet of Things (IoT) Security

The study [101] presents an advanced approach for improving multimodal biometric identification by integrating facial and iris recognition using deep learning techniques. The WVU-Multimodal dataset is employed for evaluation, and multiple Convolutional Neural Network (CNN) architectures—ResNet-50, InceptionNet, XceptionNet, and a fine-tuned CNN—are utilized to extract discriminative features from biometric images. To enhance fusion efficiency, the study investigates both early fusion, which combines raw face and iris images at the input level, and score-level fusion methods. Image quality is improved using a Gabor-based enhancement technique that modifies histogram equalization locally, thereby facilitating more effective feature extraction. The system's performance is evaluated using

standard biometric metrics, including accuracy, EER, and Receiver Operating Characteristic (ROC) curves. A comparative analysis is conducted to evaluate the effectiveness of the CNN architectures and fusion strategies, confirming the promise of CNN-based multimodal fusion in achieving higher accuracy and robustness in biometric authentication systems.

These applications highlight the synergy between multibiometric systems and soft computing, establishing a new standard for secure, adaptive, and user-friendly authentication mechanisms.

5 Analysis and Discussion

This section presents a structured analysis of the reviewed literature on soft computing techniques in biometric authentication. Table 5 first provides a comparative overview of the principal soft computing methods examined in this review, evaluated across four dimensions: reported accuracy, computational cost, scalability, and robustness to noise. This cross-method comparison is followed by Tables 6 and 7, which present detailed summaries of individual studies employing machine learning and fuzzy-based approaches respectively.

It is important to acknowledge that the comparative values presented in Table 5 are drawn from studies conducted under heterogeneous experimental conditions. Differences in benchmark datasets (varying subject counts, acquisition environments, and demographic profiles), evaluation protocols (closed-set vs. open-set identification, k-fold cross-validation vs. held-out test sets), and biometric modalities preclude direct quantitative comparison between methods. The reported accuracy figures should therefore be interpreted as indicative of performance trends within each method's primary application context rather than as absolute cross-study rankings. Standardized evaluation benchmarks and shared experimental protocols remain essential for enabling more rigorous assessment of the relative strengths and limitations of soft computing approaches across biometric modalities. Consequently, the comparative analysis in Table 5 should be interpreted primarily as a qualitative synthesis of reported trends rather than a definitive ranking of algorithms.

The comparative analysis reveals three consistent patterns. First, deep learning architectures achieve the highest accuracy but impose prohibitive computational demands for resource-constrained deployments. Second, fuzzy logic and fuzzy–neural hybrid systems, while yielding lower standalone accuracy, offer unique robustness advantages under uncertain or noisy biometric inputs. Third, swarm intelligence methods—particularly PSO and its hybrids—function effectively as optimization layers, delivering consistent accuracy improvements when integrated with classifiers. Detailed study-level evidence supporting these patterns is presented in Tables 6 and 7 below.

An overview of numerous thorough approaches suggested in biometric research is given in Table 6. These

Table 5 Comparative analysis of soft computing methods applied to biometric authentication systems

Method	Ref	Dataset / Evaluation Protocol	Reported Accuracy	Computational Cost	Scalability	Robustness to Noise	Primary Biometric Application
SVM	[105]	14-subject ECG dataset (BIOPAC); 10-fold cross-validation; unimodal	98.7%	Moderate	Limited (large N)	Moderate	ECG, fingerprint, face
k-NN	[95]	FEI face + IITD palmprint; closed-set identification; multimodal score fusion	– (fusion)	Low (train) / High (test)	Poor	Low	Multimodal: face + palmprint
ANN	[139]	30 users, 3 smartphones; held-out test; voice unimodal	>98.29%	Moderate	Moderate	Moderate	Voice authentication
CNN / Deep Learning	[101]	WVU-Multimodal dataset; score-level fusion; face + iris multimodal	>99%	High	High (GPU)	High	Multimodal: face + iris
Fuzzy Logic	[44]	Standard fingerprint + iris benchmarks; unimodal and multimodal; cross-validation	96%	Low	Moderate	Very High	Iris, fingerprint fusion
PSO	[111]	2 EEG datasets (13 and 109 subjects); closed-set identification; unimodal	99%*	Moderate	High	Moderate	EEG biometric identification
Genetic Algorithm	[44]	Standard fingerprint + iris benchmarks; hybrid unimodal/multimodal	96% (hybrid)*	High	Moderate	High	Feature selection, classifier tuning
ACO	[112]	Multimodal biometric dataset; score-level fusion; face + fingerprint	– (fusion)*	Moderate	Moderate	Moderate	Score-level fusion optimisation
Fuzzy–Neural Hybrid	[140]	Hand tremor dataset; BAN Logic security verification; unimodal	High*	High	Moderate	Very High	Metaverse biometric authentication
PSO–CNN Hybrid	[138]	Multimodal voice + iris dataset; held-out test; score-level fusion	>98%	Very High	High	High	Multimodal: voice + iris

(*) for optimisation methods, values reflect system accuracy after applying the technique for feature selection or parameter tuning

Table 6 Summary of comprehensive methods proposed for biometric research

Ref	Year	Methods	Biometric	Dataset	Results
[141]	2020	A 256-bit private key is generated by applying the SHA256 hashing algorithm to a binary string, which is subsequently used as input to the Diffie-Hellman (DH) methodology to compute the corresponding public key. For fingerprint processing, the pair-minutiae feature extraction method is utilized to derive relevant biometric features.	Fingerprint	FVC 2004, FVC2002, and NIST special database IV	it offers long-term protection for messages sent back and forth between two users.
[142]	2016	Delaunay triangulation convolution code	Fingerprint	FVC 2002 DB1, DB2, DB3	FVC 2002 DB3, EER = 6.87, FVC 2002 DB1, EER = 1.66, FVC 2002 DB2, EER = 1.89.
[143]	2015	They use two functions (hash code and parity code). The hash code is dispersed using a single-cell array. There are chaff points in another cell array. Now, a regenerative XorCoding tangles parity code.	Iris, palmprint	PolyU, IITD palm, IITD iris	When FAR = 0, FRR = 0.91
[144]	2014	Create a hybrid system with features from palmprints and fingerprints. This approach combines a 2^N discretization mechanism with random tilling.	Palmprint, Fingerprint	FVC 2004 DB1, PolyU, FVC 2002 DB1	When $d^1 = 10.51$, EER = 0.
[145]	2016	For minutiae-based fingerprint biometrics, an "ECC-free key binding scheme along with cancelable transforms is proposed." A powerful non-invertible cancelable transform known as modified Randomized Graph-based Hamming Embedding (RGHE) protects the sensitive data.	Fingerprint	FVC 2004 DB2, FVC 2002 DB1 DB2, DB3, DB4	-
[25]	2022	Develop an efficient algorithm using Fourier decomposition method (FDM) and phase transform (PT)	Electrocardiogram (ECG) signals	MIT-BIH, ECG-ID and Check Your Biosignals Here Initiative (CYBHi)	accuracies = 91.07% or the CYBHi, 97.92% for the MIT-BIH, and 98.45% for the ECG-ID
[146]	2018	proposed a method that uses a sequence of gestures as a password, enhanced that approach by introducing support vector machines (SVM) to improve gesture recognition accuracy.	surface electromyogram	Eight students of University of Miyazaki, DL-3100 (S&M Inc.)	
[105]	2019	empirical mode decomposition (EMD) to first denoise a single lead raw ECG data. EMD also extracts the region of interest from ECG signals that contain the most typical information relevant to the subject's identification.	Electrocardiogram (ECG)	14 different human subjects using BIOPACsystems.	accuracy = 98.7%, sensitivity = 100%, and specificity = 98.8%.
[147]	2021	recognition is carried out using blood volume pulse (BVP) and electrodermal activity (EDA), To find the best classifier and the most efficient representation of the EDA and BVP signals under consideration, several tests have been conducted. By using CNNs to extract discriminative features from the combined spectrograms	blood volume pulse (BVP), electrodermal activity (EDA)	17 subjects from commercial wristbands	average correct identification rate of 98.58%, with a duration of 30 seconds.

techniques use multiple datasets and algorithms to handle distinct biometric analysis and prediction areas.

In the initial study, the authors in [141] suggested a framework for two users to communicate securely utilizing a crypto-biometric system based on fingerprints. First, the user's fingerprint is used to compute the feature bit-string. The corresponding users' private keys are then obtained using revocable transformation. The public keys are then generated from the sender's and recipient's private keys using the Diffie–Hellman (DH) method. The long-term security of messages sent between two users is another benefit of our effort. The experimental evaluation over four FVC 2002 datasets, four FVC 2004 datasets, and the NIST special database IV clarifies that the suggested framework protects privacy and may be used for actual access control systems. In [142] research, they offer a fuzzy commitment approach for a fingerprint bio-cryptosystem that is cancelable. A Delaunay triangulation net made from fingerprint minutiae is used to alter the minutiae of a fingerprint. Convolution coding is also used to encrypt these altered features. The codeword is obtained during the decoding step using the Viterbi algorithm. Experiments conducted on FVC 2002 databases demonstrate the validity of the suggested approach. For FVC 2002 DB1, DB2, and DB3, the EERs

Table 7 A comprehensive study of the proposed fuzzy methods

Ref	Year	Methods	Datasets	Results
[44]	2020	AFGA, a hybrid soft computing-based optimization device that can be applied to both unimodal and multimodal biometric authentication systems	Fingerprint, Iris	With 96% accuracy, iris biometrics appears to be even more accurate when compared to other biometrics.
[111]	2025	combining a particle swarm optimization algorithm (PSO) with a fuzzy logic-based classification system for biometric identification utilizing EEG signals.	2 datasets of 13 and 109 people	high accuracy of 99%
[140]	2024	This study proposes a two-phase security framework that integrates CNNs and fuzzy logic for biometric authentication, combined with a lightweight cryptographic protocol to secure the Metaverse.	Hand tremor data	The results demonstrate that the proposed framework is highly accurate in user authentication and has strong defensive capabilities against common cyberattacks
[148]	2023	presented fuzzy identity-based encryption (FIBE). To ensure that only those with permission can access their PHR, FIBE uses biometric encryption mechanisms.	Fingerprints database	FIBE offers a more reliable and secure method for exchanging and storing PHRs in a cloud setting.
[149]	2022	The authentication process begins with identifying authentication keys, extracting the passcodes for the intended user, and sending dummy codes to unauthorized individuals. The passcode is then examined to confirm whether the user is authorized. Encryption is performed in multiple stages, depending on the number of users involved. During this step, valid codes are activated for fuzzy encoding of both text and image data. Huffman coding is used to compress the final code.	Fingerprint with phone's camera	the suggested hybrid coding method results in a 15.9% reduction in the volume of information.
[150]	2020	using fuzzy extractor-based techniques in the loop, this research suggests a unique two-factor biometric authentication system that enables the effective and safe integration of user biometrics.	Human fingerprints	The suggested scheme is an essential development in the field of intelligent and dependable security solutionst, it is resistant to known attacks using lightweight symmetric operations.

obtained using the suggested method are 1.66%, 1.89%, and 6.87%, respectively. In [143] an architecture for a biometric cryptosystem that uses biometric modalities to disguise a cryptographic key is presented. Two functions are used in this research to secure the candidate biometric modality: 1) BCH encoding, which provides the parity code saved for the query biometric template alignment, and 2) the Hash function, which computes hash-code to ensure its integrity. The cryptosystem is built by establishing two separate cell-arrays. Two techniques of implementation and evaluation are used for the suggested algorithm: 1) unimodal and 2) multimodal. Multimodal cryptosystems have proven superior to unimodal cryptosystems by experimental results from publicly accessible databases. In [144], they suggest fusing many biometric modalities at the feature-level to create an integrated template. They then suggest employing a hybrid template protection technique to secure the fused templates. The suggested approach consists of an equal-probability 2N discretization scheme and a feature modification technique called Random Tiling. The experimental findings demonstrate that, while maintaining template security, the suggested multibiometric template protection strategy outperforms its unibiometric equivalents in terms of verification outcomes. In [145] for minutiae-based fingerprint biometrics, they suggest cancellable transforms and an Error Correction Code (ECC-free) key binding approach. In addition, a robust non-invertible cancellable transform provides good protection for the details. A more efficient matcher can be selected since the method can be applied to different kinds of biometric features rather than just binary biometrics, which fuzzy commitment (FC) requires. Tests on FVC2004 and FVC2002 demonstrate that the accuracy performance is on par with the most advanced methods. In [25], the authors examined the intrasubject and intrasession validity of biometric identification systems based on ECG data. Using the phase transform (PT) and the Fourier decomposition method (FDM), they created an effective algorithm. The ECG signal is first separated into frames, each of which consists of one or more beats. Both intrabeat and interbeat fluctuations are captured in these shots. Using FDM, they are broken down into a collection of Fourier intrinsic band functions (FIBFs), from which pertinent features are retrieved. Furthermore, PT has been employed to draw attention to the inherent information concealed in the ECG signals' phase. The performance of the algorithm is examined concerning changes in the frame size, the decomposition levels, and the number of training and testing sessions. The proposed method was evaluated using three datasets: ECG-ID, MIT-BIH, and the Check Your Biosignals Here Initiative (CYBHi). While the MIT-BIH dataset was collected in an on-body (on-the-person) setting, the ECG-ID and CYBHi datasets were obtained in off-body (off-the-person) environments.

To perform classification, Random Forest (RF), Ensemble Subspace Discriminant (ESD), and SVM algorithms were employed. The experimental results indicate that the proposed approach outperforms most contemporary state-of-the-art techniques, achieving identification accuracies of 91. 07% in CYBHi, 97. 92% in MIT-BIH, and 98. 45% in ECG-ID. In [146], the authors investigated a new user authentication approach for mobile devices that utilizes surface electromyogram (s-EMG) signals rather than conventional touch-based inputs. s-EMG signals, measured on the skin's surface, originate from the electrical activity generated by muscle fiber contractions. By analyzing these signals, it is possible to differentiate among various muscle movements. Capitalizing on this property, the study proposed an authentication method in which a sequence of gestures functions as a password. The authors enhanced that approach by introducing support vector machines (SVM) to improve gesture recognition accuracy. To assess the effectiveness of the SVM-based approach as a gesture classifier, a number of tests were conducted. In [105], they presented a novel approach for an ECG-based biometric authentication system. The suggested method uses empirical mode decomposition (EMD) to first denoise a single lead raw ECG data. EMD also extracts the region of interest from ECG signals that contains the most typical information relevant to the subject's identification. Five features from the statistical, time, and frequency domains are then combined to conduct feature extraction. Ultimately, a variety of classification techniques were used, including SVM, DT, and KNN, to classify the chosen features. According to a classification evaluation based on 10-fold cross validation, SVM with a cubic kernel successfully classifies 14 subjects with the greatest accuracy of 98.7%, sensitivity of 100%, and specificity of 98.8%. In the study [147], the viability of biometric recognition is assessed by comparing recordings made during two independent sessions that were separated by an average of seven days, taking into account data gathered from commercial wristbands that are readily available on the market. More specifically, recognition is carried out using blood volume pulse (BVP) and electrodermal activity (EDA), taking into account measurements made from 17 participants engaging in normal activities like attending or instructing courses. To find the best classifier and the most efficient representation of the EDA and BVP signals considered, several tests have been carried out. The use of CNNs to extract features from combined EDA and BVP spectrograms resulted in excellent recognition performance, achieving an average identification accuracy of 98.58% on 30-second test samples.

In Table 7, the published literature reviews the role of fuzzy logic in biometric research and fuzzy logic in conjunction with metaheuristic algorithms. Fuzzy logic, which gives variables truth values ranging from 0 to 1, has been important for improving different optimization techniques and discovering uses in the biometric domain.

In [44], the "Adaptive Fuzzy Genetic Algorithm (AFGA)," a hybrid soft computing-based optimization device that can be applied to both unimodal and multimodal biometric authentication systems, is proposed. The outcomes generated by this device provide highly reliable verification by utilizing multi-modal biometric fusion supported by the powerful capabilities of fuzzy logic. Experiments conducted under diverse biometric data conditions reveal significant impacts on prevailing verification techniques. In [111], the study investigates the possibilities of combining a particle swarm optimization algorithm (PSO) with a fuzzy logic-based classification system for biometric identification that uses EEG signals. Two datasets of 13 and 109 people were used to evaluate the methodology, and the findings demonstrated that it permits high accuracy ratios of 99%. The study offers proof of this method's potential for real-world biometric identification applications. In [140], to address the inherent unpredictability of biometrics, this study proposes a two-phase security framework that integrates CNNs and fuzzy logic for biometric authentication, alongside a lightweight cryptographic protocol to secure the Metaverse. The CNN model is used at the first level to extract accurate biometric signatures from hand tremor data, and a cryptographic scheme is used at the second level to provide safe mutual authentication between the user and the Metaverse infrastructure. The findings, confirmed through BAN Logic analysis, show that the proposed framework delivers high user authentication accuracy and strong protection against prevalent cyberattacks. In [148], the authors introduced fuzzy identity-based encryption (FIBE) as a solution for secure storage and sharing of personal health records (PHRs) in the cloud. FIBE employs biometric encryption techniques, utilizing unique characteristics like voice patterns or fingerprints during both encryption and decryption processes to ensure that only authorized users can access their PHRs. This method enhances security and reliability in managing sensitive medical data within cloud environments.

In [149], greater security to create secure access can be ensured by biometric-based authentication. Using the phone's camera, fingerprints are scanned as a biometric factor. For identification and authentication, a few essential characteristics of the scanned fingerprint are extracted, sent to the convolutional neural network's learning system, and saved in a database. The authentication process involves identifying the authentication keys, extracting the passcodes for the user of interest, and sending zero codes to the fake individuals. Lastly, the passcode is analyzed to determine whether the user is authorized or not. Depending on

the number of members, encryption is done in stages. In this step, the legal codes for fuzzy encoding of the text and image data are enabled. Huffman coding is used to compress the final code before it is sent to the network or stored. Tests of the suggested approach demonstrate that this cascade encryption technique provides exceptional security, the suggested hybrid coding method results in a 15.9% reduction in the volume of information. In [150], By using fuzzy extractor-based techniques in the loop, this research suggests a unique two-factor biometric authentication system that enables the effective and safe integration of user biometrics with physically unclonable functions, a physical item that is comparable to a human fingerprint. Perfect forward secrecy (PFS) is made possible for protocol participants by this combination. The well-known real-or-random model is used to test the suggested scheme's security. Performance analysis demonstrates that the proposed scheme marks an important advance in intelligent and reliable security solutions. It ensures perfect forward secrecy (PFS), untraceability, and anonymity for users while remaining resistant to known attacks through lightweight symmetric operations.

5.1 Critical Synthesis

Beyond reporting performance outcomes, the studies reviewed in Tables 5, 6, and 7 reveal three consistent interpretive patterns that help explain why certain soft computing techniques succeed or fail under different biometric scenarios.

First, evolutionary algorithms—particularly PSO and genetic algorithms—demonstrate superior feature selection efficacy in biometric systems because biometric feature spaces are non-convex and high-dimensional, causing gradient-based methods to converge to local optima. PSO-based feature selection achieves 3–8% point accuracy gains over full-feature baselines across fingerprint, iris, and EEG modalities [109, 111]. The population-based global search of evolutionary methods is uniquely suited to navigating these landscapes without requiring explicit gradient information.

Second, fuzzy–neural hybrid systems consistently outperform standalone classifiers in decision-level fusion because fuzzy membership functions explicitly model the uncertainty inherent in matching scores from heterogeneous modalities. The AFGA framework [44] achieves 96% accuracy on iris and fingerprint fusion versus 88% for the best unimodal SVM baseline—a gap attributable to the fuzzy system's ability to handle cases where one modality produces an unreliable score, compensating with evidence from the complementary modality.

Third, purely deep CNN architectures—while often achieving the highest recognition accuracy on controlled benchmark datasets—frequently exhibit reduced generalization when evaluated across heterogeneous sensors, acquisition conditions, or demographic groups that were not adequately represented during training. Cross-database evaluations in the reviewed literature report performance drops of 8–15% points when models trained on one sensor type are applied to data captured by another sensor [101, 102]. This degradation is largely attributable to the tendency of deep models to learn dataset-specific acquisition artefacts, including sensor noise characteristics, illumination variations, and image quality differences. These findings highlight the importance of developing more robust cross-domain biometric models and motivate the adoption of domain adaptation, federated learning, and cross-sensor evaluation frameworks as key directions for future research [151].

6 Limitations and Challenges

Despite significant advances, six critical challenges remain:

(1) **Data Variability and Sensor Noise.**

Definition and evidence: Biometric samples are frequently affected by variations in illumination, pose, occlusion, environmental conditions, user behavior, and sensor noise. Such variability introduces significant intra-class variations that can negatively affect feature extraction and matching performance. The reviewed studies indicate that recognition accuracy often decreases when biometric data is acquired under unconstrained real-world conditions compared with controlled laboratory environments.

Current partial solutions and limitations: Existing approaches employ image enhancement, noise reduction, data augmentation, and robust feature extraction techniques to mitigate data variability. While these methods improve performance under specific conditions, they often fail to generalize across diverse acquisition environments and may require extensive parameter tuning or additional training data.

Recommended direction: Future research should focus on developing adaptive soft computing models capable of explicitly handling uncertainty and environmental variability. The creation of large-scale, demographically balanced datasets containing realistic acquisition conditions would further support the development of more robust biometric authentication systems.

(2) **Adversarial Attacks and Anti-Spoofing.**

Definition and evidence: The increasing deployment of biometric authentication systems has been accompanied

by growing vulnerability to presentation attacks, including replay attacks, synthetic fingerprints, printed facial images, and deepfake-based impersonation. Several reviewed studies demonstrate that high recognition accuracy under benchmark conditions does not necessarily imply robustness against sophisticated spoofing attempts.

Current partial solutions and limitations: Current countermeasures include presentation attack detection (PAD), liveness detection techniques, adversarial training, and multimodal biometric fusion. Although these methods improve resilience against known attacks, many solutions exhibit limited generalization when confronted with previously unseen attack strategies and often increase computational complexity.

Recommended direction: Future work should investigate adaptive soft computing frameworks that jointly optimize authentication accuracy and attack resistance. Standardized adversarial evaluation protocols and uncertainty-aware hybrid architectures may contribute to improving robustness against evolving spoofing techniques.

(3) **Privacy and Ethical Concerns.**

Definition and evidence: Unlike passwords or authentication tokens, biometric characteristics are permanent and cannot be replaced once compromised. The reviewed literature highlights concerns related to biometric template leakage, unauthorized data sharing, demographic bias, user consent, and compliance with emerging privacy regulations.

Current partial solutions and limitations: Several privacy-preserving approaches have been proposed, including cancelable biometrics, template protection schemes, homomorphic encryption, secure multiparty computation, and federated learning. However, these techniques frequently introduce additional computational overhead or communication costs and may negatively affect recognition performance.

Recommended direction: Future research should focus on privacy-preserving multimodal biometric frameworks that simultaneously address authentication performance, fairness, transparency, and regulatory compliance. Standardized evaluation methodologies for privacy, fairness, and security are also needed.

(4) **IoT and Edge Deployment Constraints.**

Definition and evidence: The growing adoption of biometric authentication within IoT ecosystems introduces significant resource constraints related to memory, processing capability, energy consumption, and communication bandwidth. Many state-of-the-art deep learning and hybrid soft computing models remain unsuitable for deployment on resource-constrained devices.

Current partial solutions and limitations: Researchers have explored model compression, pruning, quantization, lightweight neural architectures, and edge-cloud collaborative frameworks. While these techniques reduce resource requirements, they often involve trade-offs between recognition accuracy, latency, and energy efficiency.

Recommended direction: Future studies should focus on lightweight biometric architectures specifically designed for edge environments. Evolutionary and swarm-based optimization algorithms may provide effective mechanisms for automatically identifying compact yet accurate model configurations suitable for real-time deployment.

(5) **Explainability.**

Definition and evidence: Most high-performing biometric authentication systems rely on complex deep learning models that operate as black boxes. The reviewed literature indicates that the lack of transparent decision-making mechanisms limits user trust, regulatory acceptance, and practical deployment in security-critical applications. Tucci et al. [152] identify the absence of biometric-specific explainability benchmarks as a major obstacle to the adoption of explainable artificial intelligence in biometric systems.

Current partial solutions and limitations: General-purpose explainability techniques such as Grad-CAM, SHAP, and LIME have been applied in related domains; however, their use in biometric authentication remains limited. Existing approaches often provide only coarse explanations and are not specifically designed to interpret verification and matching decisions.

Recommended direction: Future research should develop biometric-specific explainability frameworks capable of identifying the biometric features contributing most strongly to authentication decisions. Such approaches could improve transparency, facilitate auditing, and support the detection of suspicious or manipulated biometric samples.

(6) **Cross-Sensor Generalization.**

Definition and evidence: A persistent challenge in biometric authentication is the performance degradation observed when systems trained using data from one sensor are evaluated on data acquired using different sensors. Variations in sensor technology, image resolution, acquisition protocols, and environmental conditions often result in significant reductions in recognition accuracy.

Current partial solutions and limitations: Domain adaptation, transfer learning, and cross-sensor calibration techniques have been investigated to reduce sensor-induced

discrepancies. Nevertheless, many of these approaches require access to target-domain data during training and frequently exhibit reduced effectiveness when applied to unseen sensors.

Recommended direction: Future research should prioritize the development of sensor-independent feature representations and standardized cross-sensor evaluation benchmarks. Large-scale multi-sensor biometric datasets covering diverse acquisition conditions would facilitate more rigorous assessment of model generalization capabilities.

7 Conclusion

This review has systematically examined the role of soft computing methodologies in advancing biometric personal authentication systems. The evidence synthesized across 153 reviewed studies leads to several key findings. First, deep hybrid architectures—particularly CNN-based models augmented with fuzzy logic for uncertainty-tolerant decision fusion, and PSO-optimized CNNs for adaptive feature selection—currently represent the state-of-the-art in multimodal biometric recognition, consistently achieving accuracy rates exceeding 98% on standard benchmarks. Second, swarm intelligence algorithms, especially PSO and its hybrids, have demonstrated consistent and reproducible gains of 3–8% points in accuracy over baseline feature sets in feature selection tasks across fingerprint, iris, and face modalities. Third, fuzzy logic, while delivering lower standalone accuracy than deep learning approaches, remains uniquely valuable in scenarios requiring robustness to uncertain, noisy, or incomplete biometric inputs—a critical advantage in real-world deployment environments.

Despite these advances, three critical research gaps persist. (i) Cross-sensor generalization: most reviewed models achieve high accuracy within a single database and sensor type but fail to maintain performance when evaluated across heterogeneous capture conditions. (ii) Explainability: the overwhelming majority of high-performing systems are black-box deep learning models, whose decision logic cannot be audited or trusted in high-stakes authentication contexts. (iii) Edge deployment: current state-of-the-art models carry computational requirements that preclude real-time deployment on mobile or IoT devices.

These gaps suggest the following five specific future research directions:

- **(a) Explainable AI (XAI) for biometric authentication:** developing gradient-based attribution methods (e.g., Grad-CAM, SHAP) adapted for biometric modalities to enable transparent, auditable authentication decisions.
- **(b) Federated learning for privacy-preserving multimodal training:** enabling collaborative model training across distributed biometric repositories without centralizing sensitive biometric templates—a critical requirement for compliance with privacy regulations such as GDPR.
- **(c) Lightweight hybrid models for edge deployment:** designing knowledge-distilled or quantized versions of hybrid soft computing architectures (e.g., compressed PSO-CNN or Fuzzy-MobileNet models) suitable for real-time inference on embedded IoT devices with constrained computational resources.
- **(d) Adversarial robustness testing:** systematically evaluating soft computing-based biometric systems against presentation attacks (PAD), deepfake-generated biometrics, and replay attacks across all major modalities.
- **(e) Standardized cross-sensor multimodal benchmarks:** developing and releasing large-scale, demographically balanced, multi-sensor biometric databases that enable reproducible cross-system comparison—a prerequisite for scientific progress in the field.

In summary, soft computing—particularly in its hybrid deep-learning and fuzzy-evolutionary forms—offers compelling and proven solutions to the core challenges of biometric authentication. Continued interdisciplinary collaboration across machine learning, security engineering, and human factors research is essential to translate these academic advances into trustworthy, scalable, and privacy-respecting real-world biometric systems.

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Declarations

Compliance with Ethical Standards This article does not contain studies with human participants or animals carried out by any author.

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